

The Impact of a Higher Cost of Credit on Exporters: Evidence from a Change in Banking Regulation*

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Abstract

How do exporters respond to changes in the cost of credit? We exploit an exogenous shift in banking regulation that increased the cost of financing for exports in the European Union. Using customs, firm-level, and credit registry data on Portuguese firms, we find that a rise in financing costs reduces exports by 8 percent through the intensive margin, while firm entry into new destinations declines. Within firms, exporters adjust their product mix, skewing it toward goods with lower working capital requirements. We provide direct evidence of the mechanism: loan rates for exporters increase, and loan amounts decline by 7 percent. We also provide evidence of credit rationing. A multi-sector Ricardian model shows that these disruptions lead to a depreciation in terms of trade, reducing welfare in both the EU and its trading partners.

Keywords: Trade Finance, Product Mix, Lending Channel

JEL Codes: G22, G30, F10, J23

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1 Introduction

How does the cost of credit shape international trade? Financial constraints are known to influence firm production (Khwaja and Mian, 2008; Chodorow-Reich, 2014; Benmelech et al., 2016), but exporters—due to their longer production cycles—may be particularly sensitive to fluctuations in credit conditions. The empirical literature on trade and finance (Manova, 2013; Paravisini et al., 2015) suggests that credit shocks primarily affect exports through the intensive margin, impacting the volume of shipments from existing exporters. However, this conclusion contrasts with theoretical predictions. In models of international trade, such as the Melitz framework, firm entry and exit play a central role in shaping trade flows, implying that credit shocks should also operate through the extensive margin. Moreover, firms frequently adjust their product mix in response to shocks (Mayer et al., 2014), which can have lasting effects on their growth and exposure to risk. These additional margins of adjustment suggest that the effects of credit shocks on trade may be deeper and more persistent than previously documented, as firms will need additional time and resources to recover from the shock. This paper examines how an increase in the cost of credit affects exports through three distinct channels: the intensive margin, the extensive margin, and adjustments in firms’ product mix.

Exporters rely more heavily on credit than firms operating solely in domestic markets. Long lags between production, shipment, and delivery create significant liquidity needs. For instance, within the European Union, a product may take up to a month to transit from the factory to a shipping vessel (Djankov et al., 2010). During these delays, exporters must finance supplier payments, increasing their reliance on working capital, which credit helps to bridge. In addition to liquidity constraints, exporters face counterparty risk: securing payment from foreign buyers can be challenging, particularly in cases of default. Banks play a crucial role in mitigating this friction by acting as intermediaries. With established relationships and affiliations with foreign banks, they are better positioned to enforce payments and manage counterparty risk efficiently. Moreover, the role of banks as intermediaries is even more pronounced when either the exporter or the importer is in a country with weak contractual enforcement (Antràs and Foley, 2015).

The predominant form of credit in international trade is trade finance. At its core, trade finance consists of working capital loans granted by domestic banks and backed by letters of credit issued by foreign banks. In this paper, we examine the impact of an increase in the cost of trade finance for certain destinations while holding costs fixed for all others. This variation arises from a change in macroprudential regulation following the implementation of Basel III.

Basel III introduces substantial changes to banks' risk management practices. Under the new regulation, banks must maintain a minimum level of equity relative to risk-weighted assets.¹ Higher-risk assets receive higher risk weights, requiring banks to hold more equity to meet capital requirements. A key feature of Basel III is the revision of risk weights for trade finance. The regulation assigns higher risk weights to banks in non-OECD countries (classified as high-risk), while leaving risk weights unchanged for banks in OECD countries (low-risk). Consequently, the cost of providing trade finance loans increases for transactions involving high-risk countries. Since banks are likely to pass part of this cost to exporters, the marginal cost of exporting to high-risk destinations rises.

We use a unique dataset on Portuguese firms to examine the impact of higher trade finance costs. Portugal provides an ideal setting for this analysis, as most of its exporters are small, financially constrained, and rely exclusively on bank credit to finance working capital. Additionally, a significant share of Portuguese exports is directed to high-risk countries, making them particularly exposed to the regulatory shock. Our dataset includes detailed information on exports by destination and product for the universe of Portuguese exporters. We complement these data with credit registry records, which contain all loans obtained by Portuguese firms, along with information on loan amounts, maturities, and interest rates. This rich dataset allows us to conduct a within-firm analysis, comparing the evolution of exports to high-risk destinations with those to low-risk destinations in response to the increase in the cost of credit.

At the firm level, we find that exports to high-risk destinations decline by approximately 8 percent relative to low-risk destinations. At the aggregate level, Portuguese exports also decrease, suggesting that firms are unable to fully compensate for lost sales in high-risk markets by expanding exports to low-risk destinations. We also find that firm entry into high-risk destinations declines by 5 percent, while exit rates remain unchanged. This asymmetry is consistent with the presence of sunk entry costs, as documented by [Alessandria and Choi \(2007\)](#). Given these costs, firms already operating in a market may continue exporting despite a decline in sales, whereas prospective entrants may forgo entry due to higher financing costs.

We next examine how exporters adjust their product mix in response to a higher cost of credit. To do so, we construct a novel classification of products based on their dependence on working capital and bank credit. This measure is derived from the product's cash conversion cycle, which captures the time required to convert inventory investments into cash from sales. At the firm level, we find that the decline in exports to high-risk des-

¹Risk-weighted assets represent the sum of all assets on a bank's balance sheet, each adjusted by an assigned risk weight reflecting its credit risk.

tinations is driven entirely by products with high dependence on bank credit. This result supports our proposed mechanism, which operates through the working capital channel, and highlights how firms adjust their product mix in response to changes in credit costs. Specifically, when the cost of credit rises, firms shift their product mix toward goods that require less working capital and bank financing, thereby reducing their costs.

We also examine the impact of the Basel III-induced increase in credit costs on loan rates and amounts obtained by exporters. To do so, we classify firms based on their exposure to high-risk destinations, measured as the share of exports to these markets prior to the implementation of Basel III. We find that exporters with high exposure to high-risk destinations experience a 13-basis-point increase in loan interest rates relative to low-exposure firms. However, this effect is small compared to the decline in exports, due to two additional margins of adjustment that introduce a downward bias in our estimate of interest rate effects. First, exporters with high exposure reduce their loan amounts, cutting borrowing by 7 percent relative to low-exposure firms. Second, these exporters also reduce their reliance on credit via the extensive margin: the probability of obtaining a loan declines by 7 percent compared to firms with lower exposure to high-risk destinations.

To assess the welfare implications of the trade disruptions induced by higher credit costs, we develop a structural model. The model not only quantifies the costs of Basel III on international trade but also examines the extent to which exporters can offset the decline in exports to high-risk destinations by increasing exports to low-risk destinations. We adopt a Ricardian trade model with multiple countries and sectors, following [Caliendo and Parro \(2015\)](#). In this framework, trade arises from cross-country differences in productivity, with firms importing goods from countries with a comparative advantage in their production. We introduce financial frictions in a tractable manner by assuming that exporters must finance production costs upfront and therefore rely on bank credit. We then simulate an interest rate shock, calibrated to match the causal effect of Basel III on EU exports. Since the model abstracts from credit market frictions, this calibration yields an estimate of the *shadow cost* of Basel III on trade—an important statistic for evaluating the policy’s impact. Our results suggest that the observed decline in exports to high-risk destinations in the EU is consistent with an interest rate increase of 1.8 percentage points. This estimate differs significantly from the effect we document for Portuguese exporters, where our empirical estimation is subject to a downward bias.

Our model estimates that welfare in high-risk countries declines by 0.04 percent as the cost of imports rises. Higher import prices increase marginal costs, eroding these countries’ comparative advantage. EU countries experience an even larger welfare loss of 0.09 percent. To contextualize this magnitude, [Caliendo and Parro \(2015\)](#) find that the total

welfare gains from NAFTA for the U.S. amount to approximately 0.06 percent, suggesting that Basel III leads to a quantitatively meaningful decline in welfare. Most of this welfare loss is driven by a deterioration in terms of trade—the ratio of export prices to import prices. As the cost of credit rises, global demand for EU exports declines due to higher prices, leading to a fall in domestic factor prices. Lower factor prices reduce firms' marginal costs, ultimately decreasing export prices. As EU exports become cheaper, welfare declines further, as EU countries can now afford fewer imports for a given quantity of exports.

Our model indicates that the increase in the cost of capital following the implementation of Basel III operates similarly to an economy-wide decline in productivity. Prior to the regulatory change, EU exporters benefited from access to relatively cheap credit, which facilitated exports to developing countries. Basel III erodes this advantage by raising financing costs, making EU exports less competitive. For developing countries, where competitiveness is partly driven by low wages, Basel III further increases marginal costs by raising the price of imported inputs. This additional cost pressure weakens their ability to compete in global markets.

The broader welfare implications of Basel III extend beyond the scope of this paper. Our analysis focuses exclusively on the costs arising from its impact on international trade. However, Basel III was introduced under the premise that banks were underpricing loans to exporters, leading to excessively low interest rates. From this perspective, higher interest rates may generate welfare gains by correcting this mispricing. These potential gains must, however, be weighed against the welfare costs we estimate.

Related Literature. Our paper contributes to a growing literature focusing on the role of credit in international trade.² Some papers, like [Gertler and Rogoff \(1990\)](#), [Matsuyama \(2005\)](#), and [Chor \(2010\)](#), have highlighted that credit is an important source of comparative advantage.³ In this literature, [Manova \(2013\)](#) shows that financial frictions substantially drag international trade. In the Great Recession, [Chor and Manova \(2012\)](#) and [Ahn et al. \(2011\)](#) show that credit played a crucial role in explaining the drop in trade flows. Other papers have focused on how financial frictions affect the impact of other shocks – [Caggese and Cuñat \(2013\)](#) study trade liberalizations, [Manova \(2008\)](#) studies credit lib-

²There is also a broader literature focusing on the effects of credit on real activity, focusing on the firm side, like in [Paravisini \(2008\)](#); [Khwaja and Mian \(2008\)](#); [Chodorow-Reich \(2014\)](#); [Huber \(2018\)](#) or [Benmelech et al. \(2016\)](#). However, exporters are usually more sensitive to credit shocks, as shown by [Minetti and Zhu \(2011\)](#).

³There is also a literature studying trade finance, which is the most common form of financing for exporters. Some examples include [Schmidt-Eisenlohr \(2013\)](#), [Antràs and Foley \(2015\)](#), and [Niepmann and Schmidt-Eisenlohr \(2017\)](#).

eralizations, [Chaney \(2016\)](#) studies currency appreciations, and [Becker et al. \(2013\)](#) study financial development. Using historical data, [Xu \(2022\)](#) finds that countries more exposed to bank failures observe a permanent export decline.

The paper most closely related to ours is [Paravisini et al. \(2015\)](#), who use the 2008 Peruvian sudden stop as a shock to credit to exporters. They find that this credit shock only affects exports through the intensive margin. We contribute to this literature in two ways. First, we find that the extensive margin and the product mix are essential to understanding the effects of credit shocks on exports.⁴ Second, the nature of our shock allows us to speak to the nature of substitution across destinations and products *within* a firm. Therefore, we can show how firms change their activities in one segment in response to a shock in another, as in [Giroud and Mueller \(2019\)](#).

We also contribute to a literature that studies the design of macroprudential policy and its impact on real activity. [Kashyap et al. \(2004\)](#), [Hanson et al. \(2011\)](#), [Kashyap et al. \(2011\)](#), [Repullo and Suarez \(2013\)](#), and [Bahaj and Malherbe \(2020\)](#) study the optimal design of banking regulation. On the empirical side, [Gropp et al. \(2019\)](#), [Aiyar et al. \(2014\)](#) show that banks respond to risk-weight increases by reducing lending.

Outline. The rest of the paper proceeds as follows. Section 2 describes the institutional framework and the changes introduced by Basel III. Section 3 describes the data and presents summary statistics. Section 4 discusses the empirical strategy and presents the results for the comparison between countries. Section 5 presents results for the effect on lending conditions. Section 6 introduces a structural model. Section 7 concludes.

2 Trade Finance and Basel III

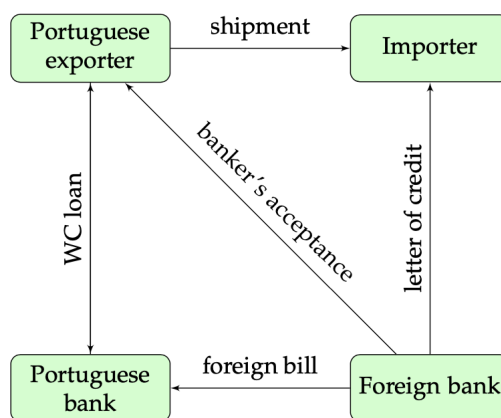
Trade finance is one of the oldest forms of credit in international finance, dating back at least to the 19th century. Its primary function is to mitigate two fundamental challenges in international trade: (1) international transactions take significantly longer to complete than domestic ones, and (2) exporters and importers often have limited legal recourse in the event of default. Empirical evidence underscores these frictions. Using data from manufacturing firms in 180 countries, [Djankov et al. \(2010\)](#) report a median delay of 21

⁴There is evidence of how exporters change their product mix in response to shocks. In the theoretical literature, [Mayer et al. \(2014\)](#) study how changes in the competitive environment lead exporters to change their product mix. [De Loecker \(2011\)](#) argues that changes in the product mix are an essential part of a firm's response to trade shocks. Using data from wine producers, [Mendes \(2020\)](#) shows that firms adjust to adverse credit shocks by skewing their product mix towards low-quality wines. In response to adverse demand shocks, [Friedrich and Zator \(2022\)](#) show that low-leverage exporters expand their product mix.

days between production and shipment. Similarly, [Hummels and Schaur \(2013\)](#) document that goods imported into the U.S. by sea typically spend 20 days in transit. Moreover, as highlighted by [Amiti and Weinstein \(2011\)](#), importers frequently defer payment for up to 90 days after receiving the shipment.

At its core, trade finance involves four key players, as illustrated in Figure 1. In the domestic country—Portugal, in our example—there is an exporter and a domestic bank. The destination country hosts an importer and a foreign bank, each playing a distinct role in facilitating the transaction.

FIGURE 1: Trade Finance



The trade finance process unfolds in five stages. First, the exporter and importer negotiate a sales contract specifying key transaction details, such as price, volume, payment terms, and delivery conditions. At this stage, the importer requests a letter of credit from the foreign bank to serve as a payment guarantee. In the second stage, the exporter secures a working capital loan from the Portuguese bank, using the letter of credit as collateral to finance production costs. The third stage involves production and shipment of goods.

In the fourth stage, the foreign bank issues a banker's acceptance to the exporter, which guarantees future payment, typically maturing in approximately 90 days. The exporter then sells the banker's acceptance to the Portuguese bank, usually at a discount, effectively transferring the payment risk. Consequently, the Portuguese bank replaces its claim on the exporter with a claim on the foreign bank. In the fifth and final stage, the foreign bank settles the banker's acceptance with a foreign bill, completing the transaction. However, default can occur at any stage of this process.

At the outset, the Portuguese bank holds a claim on the exporter, similar to a conventional working capital loan extended to a domestic firm. Over time, however, this claim

shifts to a foreign bank, introducing an element of off-balance-sheet risk. Unlike standard domestic loans, repayment ultimately depends not only on the exporter but also on the foreign bank's ability to fulfill its obligations.⁵

2.1 Treatment of trade finance under Basel III

Basel III, approved in all E.U. countries in 2013 and implemented on January 1, 2014, is a globally agreed-upon regulatory framework developed by the Basel Committee on Banking Supervision in response to the 2007–2009 Global Financial Crisis.⁶ The framework introduced three major reforms: (1) stricter capital requirements, including revised methodologies for calculating capital ratios, (2) the implementation of macroprudential controls, such as capital buffers, and (3) new measures to mitigate excessive leverage and liquidity risk.

The capital ratio is given by

$$\text{Capital ratio} = \frac{\text{Tier 1 capital}}{\sum_k \omega_k \text{Asset}_k}$$

where the numerator represents Tier 1 capital, consisting of equity and disclosed reserves, and the denominator represents total risk-weighted assets. The risk weight ω_k assigned to each asset k reflects its relative risk: riskier assets receive higher weights. For instance, a AAA-rated bond carries a risk weight of 0.1, while a bond rated below B is assigned a risk weight of 1.

Prior to Basel III, short-term claims on foreign banks were uniformly assigned a risk weight of 0.2, regardless of the foreign bank's country of origin or creditworthiness. Basel III introduced a more differentiated approach by tying risk weights to the credit rating of the foreign bank. Trade finance loans remain subject to a 0.2 risk weight if the foreign bank is classified as low-risk, but if the foreign bank is classified as high-risk, the risk weight increases to 0.5.⁷

⁵Trade finance also relies on alternative financial instruments. [Antràs and Foley \(2015\)](#) identify three: cash-in-advance, open account terms (which do not require direct bank intermediation), and documentary collection terms (which, like the letter of credit process, involve financial intermediation). Their findings suggest that large firms exporting to countries with strong contract enforcement tend to favor financing methods that do not require direct intermediation. In terms of scale, [Niepmann and Schmidt-Eisenlohr \(2017\)](#) use SWIFT data from 2007 to 2012 to estimate that bank-intermediated trade accounts for approximately 15 percent of global trade volume, while other estimates place this share closer to 47 percent.

⁶In the E.U., Basel III is outlined in two key legislative documents: Directive 2013/26 of June 26, 2013, which establishes the general principles, and Regulation 575/2013 of November 30, 2013, which provides a more detailed regulatory framework. These are transposed into national law by all E.U. member states.

⁷According to Articles 120 and 121 of E.U. Regulation 757/2013 (June 2013), trade finance loans involving unrated foreign banks with maturities exceeding three months receive a risk weight of 0.5. Unrated

This adjustment raises the marginal cost of trade finance loans involving high-risk foreign banks, while leaving the cost unchanged for loans involving low-risk foreign banks.

Since we do not directly observe the credit rating of foreign banks, we use the OECD's measure of sovereign risk as a proxy, classifying destinations into two groups: OECD countries are designated as low-risk, while non-OECD countries are classified as high-risk.⁸ Using sovereign ratings as a proxy for foreign bank ratings is unlikely to underestimate bank risk, as it is improbable that a poorly rated bank would operate in a low-risk country. In fact, bank ratings often exceed sovereign ratings, particularly in emerging economies.

In our empirical analysis, we compare the evolution of exports to high-risk and low-risk countries. Foreign banks in low-risk countries are consistently classified as low-risk, while some banks in high-risk countries may also receive a low-risk classification. This classification bias could attenuate the estimated effects of Basel III on exports, making our findings more conservative.

Moreover, firms may respond to rising credit costs by shifting to lower-risk banks. Firms in countries classified as high-risk may reallocate their financing relationships to banks in low-risk jurisdictions, introducing additional measurement error correlated with the regulatory shock. This reallocation would further dampen the estimated impact, as trade finance costs would remain unchanged if letters of credit were issued by low-risk banks.⁹

To gauge the potential magnitude of this regulatory shock, we consider a simple model of bank asset allocation. Suppose a bank can invest in three types of assets: a risk-free asset F , a risky assets A , and trade finance T . Total assets are given by $F + A + T$ and the bank's Tier 1 capital is E . Each asset i has a risk weight ω_i .

Suppose the bank aims to maintain a constant capital ratio, total assets, and Tier 1 capital.¹⁰ Under Basel III, the risk weight for the risk-free asset is $\omega_F = 0$. Assume further that the bank treats returns on all asset classes as exogenous. The marginal cost of trade finance is the opportunity cost of increasing trade finance T by one unit, which corresponds to the forgone return on the bank's remaining assets. Given these assumptions,

institutions—typically banks lacking formal credit ratings—are generally considered high-risk. Most banks in low-income countries, for example, fall into this category (BIS, 2015).

⁸The OECD publishes sovereign ratings for most countries, which banks use to determine the appropriate risk weights for exposures to foreign banks.

⁹If firms successfully switch to low-risk banks, the expected increase in trade finance costs would not materialize, reducing the measured impact of Basel III on exports.

¹⁰Empirical evidence suggests that banks adjust to increases in capital requirements by reducing risk-weighted assets rather than raising equity. For instance, Gropp et al. (2019) show that banks respond to higher capital ratios by cutting lending to corporate and retail clients rather than increasing their capital base. In our framework, this implies a reduction in investment in the risky asset A .

the marginal cost is given by

$$\begin{aligned}\text{Marginal cost} &= r_f \times (-\Delta F) + r_A \times (-\Delta A) \\ &= r_f + \underbrace{\frac{\omega_T}{\omega_A} \times (r_A - r_F)}_{\text{risk premium}},\end{aligned}\tag{1}$$

which consists of the risk-free rate plus a risk premium that depends on the ratio of the risk weights assigned to trade finance and risky assets (relative to risk-free assets). This expression formalizes the idea that the marginal cost of trade finance is tied to divestment from risky assets.

To increase trade finance, a bank can reduce its holdings of risk-free assets while keeping total assets unchanged. However, since risk-free assets carry a zero risk weight under Basel III, this reallocation increases the bank's total risk-weighted assets. To maintain a constant capital ratio, the bank must simultaneously reduce its exposure to risky assets, which typically offer higher returns. The extent of this divestment depends on the relative risk weights of trade finance and risky assets.

Empirical evidence on how banks responded to Basel III remains limited. However, one key source is the International Chamber of Commerce (ICC), which represents over 45 million companies across more than 100 countries. In 2014, the ICC published *Rethinking Trade Finance*, a report focusing on the impact of Basel III on export finance. A survey of export finance professionals within banks revealed that 78 percent believed Basel III had increased the cost of export finance, while 69 percent reported that banks had raised prices for customers as a result.¹¹

3 Data

We construct a dataset containing detailed annual export volumes for all Portuguese firms, disaggregated by destination and product. Additionally, we incorporate credit registry data, which provides information on all loans obtained by Portuguese firms. Finally, we supplement these data sources with firm-level balance sheet information.

¹¹The ICC report also provides insight into the risk characteristics of trade finance. Using data from 24 banks covering over 4.5 million transactions—representing exposures of approximately \$2.4 trillion—they find that the customer default rate for the type of trade finance described here is just 0.033 percent. By comparison, a corporate debt instrument with a similar default rate would receive a Moody's credit rating between Aa and Aaa. This suggests that despite regulatory changes, trade finance remains a low-default-risk asset.

3.1 Data sources

Trade data. We use a dataset from Statistics Portugal, which provides monthly information on all exports and imports by Portuguese firms, disaggregated by firm, source/destination country, and product classification, covering the period from 2011 to 2018. Products are classified according to the European Combined Nomenclature, an eight-digit system used by the European Union. This classification extends the six-digit Harmonized System (HS) by adding two additional digits. In our analysis, we aggregate goods at the HS heading level, which consists of the first four digits. To align with firm-level balance sheet and income statement data, which are reported annually, we further aggregate trade data to the annual level.

Credit registry data. We use the Portuguese credit registry, *Informação Individual de Taxas de Juro*, managed by the Bank of Portugal, which contains data on all new loans and loan renegotiations between 2013 and 2018. Until December 2014, only banks with an annual volume of new loans to firms exceeding €50 million were required to report loan details. From January 2015 onward, this reporting obligation was extended to all banks. For each loan, we observe the origination date, bank, firm, interest rate, maturity, loan amount, and loan type. However, trade finance loans cannot be directly identified in this dataset. We exclude financial firms, as well as overdrafts and renegotiated loans, to focus on new lending. To complement the credit data, we incorporate firm-level balance sheet and income statement information from *Informação Empresarial Simplificada*, a joint initiative of the Ministry of Justice, the Ministry of Finance, Statistics Portugal, and the Bank of Portugal. Since all Portuguese firms are required to report this information, the dataset represents the full universe of firms in the country.

Merged data. Due to legal restrictions imposed by the Bank of Portugal and Statistics Portugal, we cannot directly merge the credit registry with the trade dataset.¹² However, both datasets contain information on total exports, though sourced differently. To link firms across datasets, we employ a matching procedure that pairs each firm in the credit registry with a firm in the trade dataset operating in the same sector (defined at the finest available level: five-digit classification) and minimizing the absolute difference in export values. This approach represents the best possible matching method given legal

¹²The credit registry is anonymized, whereas the trade dataset is not. Merging them directly would compromise anonymization.

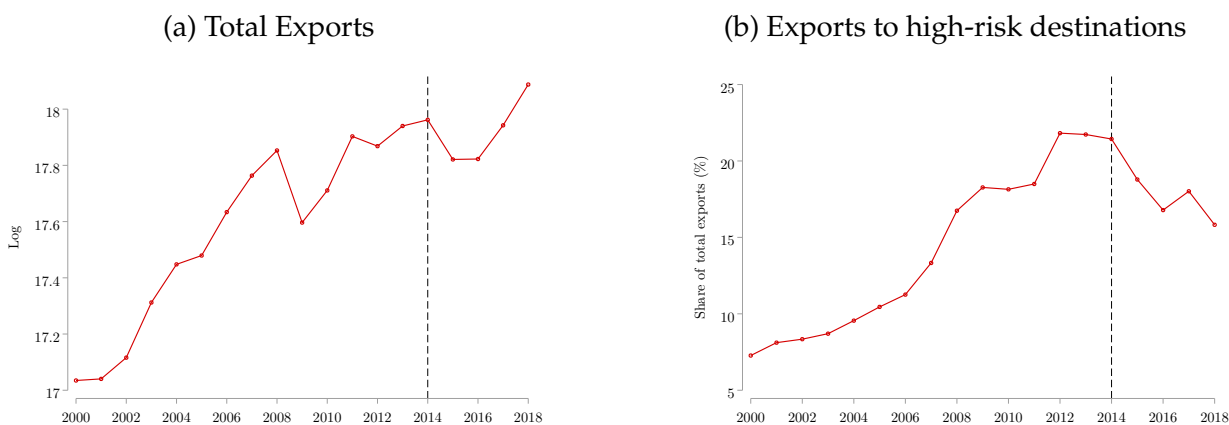
constraints. Through this process, we identify 11,246 exporters in the credit registry.¹³ These firms are matched to 6,453 exporters in the trade dataset. Among these, 4,079 firms achieve unique matches, meaning each firm in the trade dataset is linked to a single firm in the credit registry.

3.2 Summary statistics

In 2013, Portuguese exports of goods totaled €47 billion, equivalent to approximately 27 percent of GDP. That year, Portugal exported to 189 destinations, with the European Union as its primary export market. Spain was the largest trade partner, accounting for 24 percent of total goods exports, followed by France (12 percent) and Germany (12 percent). Angola was the fourth-largest trade partner, representing 7 percent of total exports. Exports to high-risk destinations made up 23 percent of total exports in 2013. The evolution of Portuguese exports is depicted in Figure 2.

FIGURE 2: Evolution of Portuguese exports

This Figure presents the evolution of Portuguese exports from 2000 to 2018, using data from CEPII. In Panel (a), we show the evolution of total Portuguese exports of goods at current prices. In Panel (b), we present the share of Portuguese exports going to high-risk countries.



Panel (a) reveals four distinct periods in the evolution of Portuguese exports. Between 2000 and 2008, exports grew substantially, increasing by approximately 80 percent. This was followed by a sharp decline during the Great Recession and the Sovereign Debt Crisis (2008–2011), after which exports quickly rebounded. The decline is commonly attributed

¹³We define exporters as firms whose annual exports constitute at least 5 percent of total sales, following Niepmann and Schmidt-Eisenlohr (2017). Relaxing this assumption does not qualitatively change our results, though increasing the threshold reduces the number of exporters, potentially weakening the statistical power of our analysis in Section 4.

to a contraction in credit supply, which contributed to the so-called Great Trade Collapse (Chor and Manova, 2012). From the end of the financial crisis until 2013, exports exhibited only modest growth. After 2013, exports experienced another decline, followed by a rapid recovery.

Panel (b) plots the share of Portuguese exports directed to high-risk countries, showing a substantial increase in this share until 2013. Between 2000 and 2013, the share more than doubled, indicating that exports to high-risk destinations grew faster than those to low-risk destinations. Notably, during the Sovereign Debt Crisis, this share accelerated as Portuguese exports to high-risk countries surged. This shift was driven by declining domestic demand and weakening demand across the EU.¹⁴ After 2014, coinciding with the implementation of Basel III, the share of exports to high-risk destinations exhibits a sharp and persistent decline.

TABLE I: Summary Statistics

This table presents summary statistics for our sample, for 2013. For each variable, we compute the mean, median, standard deviation, minimum and maximum across all firms. We present summary statistics for the number of destination to which a firm exports, the number of products a firm exports, the average (across destinations) number of products per destination, the share of the firm's main destination in its exports, the share of the firm's best selling product in its exports, the HHI across destinations, the HHI across products, exports to high-risk destinations as a share of total exports, the number of loans obtained, the number of banks from which a firm obtain loans, and loan maturity (in days).

	Mean	Median	SD	Min	Max	N
Number of destinations	4.66	2.00	7.23	1.00	84.00	11,159
Number of products	14.67	5.00	27.69	1.00	342.00	11,159
Average number of products per destination	11.23	3.00	22.92	1.00	294.00	11,159
Share of main destination	0.45	0.11	0.48	0.00	1.00	11,159
Share of main product	0.24	0.00	0.41	0.00	1.00	11,159
HHI for destinations	0.76	0.94	0.28	0.05	1.00	11,159
HHI for products	0.65	0.69	0.32	0.02	1.00	11,159
Share of exports to high-risk destinations	0.71	1.00	0.41	0.00	1.00	8,331
Number of loans	15.72	4.00	46.28	1.00	1,717.00	11,159
Number of banks	1.72	1.00	1.15	1.00	9.00	11,159
Loan maturity (in days)	371	92	661	1	6,995	11,159

¹⁴As documented by Almunia et al. (2021), exporters in Southern Europe responded to domestic demand contractions by increasing exports. However, given the strong synchronization of business cycles within the EU, this export expansion was directed primarily toward emerging, high-risk markets.

Table I presents summary statistics on Portuguese exporters in our sample. The average exporter sells 15 products across five destinations, with an average of 11 products per destination. However, the distribution is highly right-skewed, as indicated by median values that are substantially lower than the means. This suggests that a small number of firms operate in many destinations and sell a wide range of products.

On average, a firm’s main product—defined as the best-selling product across all destinations—accounts for 45 percent of its total exports, while its primary export destination represents 24 percent. The Herfindahl-Hirschman Index (HHI) further confirms this concentration, showing that exports are highly concentrated both across destinations and products.

On average, 71 percent of a firm’s exports are directed to high-risk destinations, though the distribution is effectively bimodal, with many firms exporting either almost exclusively to low-risk or to high-risk destinations. Given that only 23.3 percent of aggregate exports go to high-risk destinations, this pattern suggests that larger exporters (in terms of total export volume) tend to focus on low-risk markets.

Regarding credit access, the average exporter obtained 16 loans in 2013, though this figure is driven by outliers, with a median of five loans. Exporters typically sourced credit from two different banks. The median loan maturity is 134 days, aligning with expectations for trade finance, which typically ranges from 90 to 180 days.

4 Effect on Exports

4.1 Intensive Margin

To evaluate the impact of higher credit costs on export volumes, we employ a difference-in-differences design at the exporter level. Within a given exporter, the marginal cost of producing goods for high-risk destinations rises relative to that for low-risk destinations.¹⁵ We estimate the following regression:

$$\log \text{Exports}_{ipt} = \lambda_{ipt} + \mu_d + \beta X_{dt} + \gamma \times \mathbf{1}\{d \in \text{High-risk}\} \mathbf{1}\{t \geq 2014\} + u_{ipt}. \quad (2)$$

where the dependent variable is the logarithm of the export value of firm i , product p , to destination d at time t . On the right-hand side, we control for firm-product-year fixed effects, destination fixed effects, and time-varying destination characteristics.¹⁶ We include

¹⁵We classify OECD member countries as low-risk and all non-OECD countries as high-risk.

¹⁶The inclusion of firm-product-year fixed effects restricts the sample to firms that export to both high-risk and low-risk destinations, ensuring within-firm comparisons.

destination-level controls X_{dt} , which comprise log GDP, log total imports, the import-to-GDP ratio, the nominal and real exchange rates, the current account-to-GDP ratio, and the sovereign risk rating, as well as region-year fixed effects. The coefficient γ estimates the average treatment effect on exports. Standard errors are clustered at the firm level.

Discussion of Assumptions. We include firm-product-year fixed effects to account for all firm- or product-specific shocks, such as productivity changes or fluctuations in product demand. These fixed effects also ensure that our identification strategy relies on within-exporter variation to estimate the causal effects of Basel III. Additionally, we include destination fixed effects to control for iceberg costs and other unobservable, time-invariant trade frictions. To account for time-varying demand conditions in destination countries, we incorporate destination-specific controls, while region-year fixed effects absorb variation in shipping time and costs.

Our identification strategy relies on two key assumptions. First, the implementation of Basel III must be exogenous to Portuguese firms' exporting decisions. In particular, Basel III should not have been designed with the specific conditions of the Portuguese export market in mind. Since Basel III is a multinational macroprudential regulatory framework—and Portugal is not a member of the Basel Committee on Banking Supervision—it is unlikely that its provisions were influenced by the actions of Portuguese exporters.

Second, our estimates could be biased if firms anticipated the effects of Basel III and adjusted their export strategies accordingly. However, most firms in our sample are small, and it is reasonable to assume that their managers lack the sophistication to predict Basel III's impact on trade finance. The only way they could infer its effects would be through a preemptive increase in interest rates. However, such a move would require Portuguese banks to raise rates before Basel III took effect, thereby reducing their own profits.

Results. We estimate equation (2) using data from 2011 to 2018, and the results are presented in Table II.

Across all specifications, we find that exports to high-risk destinations decline relative to exports to low-risk destinations following the implementation of Basel III. In our preferred specification (column 4), our empirical analysis indicates that the increase in credit costs due to Basel III led to an almost 8 percent decline in exports to high-risk destinations relative to low-risk destinations. If this reduction is not offset by increased exports to other markets, total Portuguese exports could decrease by up to 2 percent.¹⁷

¹⁷This estimate is obtained by multiplying the coefficient by the share of exports directed to high-risk destinations in 2013.

TABLE II: Effects on volume of Portuguese exports

This Table presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time, and where compare the period between 2011 and 2013 with the period after 2014. We use annual data from 2011 to 2018. We present estimates for the average treatment effect. We include destination and firm-product-year fixed effects, and we also present a specification with destination, time, product and firm fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. Errors are clustered by firm. ***, **, * denote significant at the 1, 5 and 10 percent levels respectively.

	(1)	(2)	(3)	(4)
High-risk $\times$ Post 2014	-0.035* (0.020)	-0.037* (0.020)	-0.050* (0.029)	-0.076** (0.033)
Share of exports going to high-risk (%)	22	22	22	22
Destination FE	✓	✓	✓	✓
Year FE	✓	✓		
Firm FE	✓	✓		
Product FE	✓	✓		
Firm $\times$ Product $\times$ Year FE			✓	✓
Destination controls		✓		✓
Observations	1,663,141	1,651,477	895,117	895,042

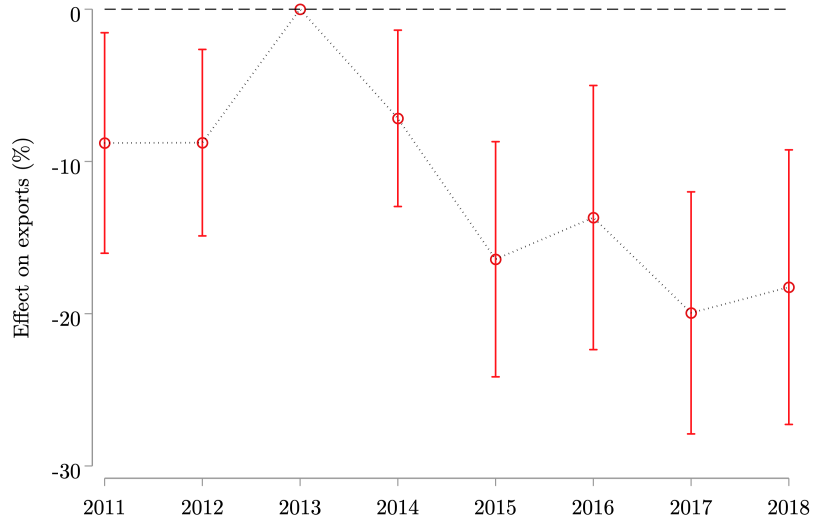
One potential concern is that Portugal's two main high-risk export destinations—Brazil and Angola—experienced economic downturns around 2013, which could bias our results. To address this, we re-estimate equation (2), including either a past colonial link indicator or a common language indicator interacted with time fixed effects to absorb variation from Portuguese-speaking countries. The results, presented in Table A.I, confirm that the average treatment effect remains negative and significant.

Additionally, we allow the average treatment effect to vary over time. The results of this analysis are presented in Figure 3.

Following 2013, we observe a persistent decline in exports to high-risk destinations, consistent with an increase in the cost of funding exports to these markets. However, the decline is not immediate: exports to high-risk destinations fall modestly in 2014, followed by a more pronounced drop in 2015, which then persists through 2018. Importantly, this comparison also reveals pre-trends, as exports to high-risk destinations did not evolve in parallel with exports to low-risk destinations before 2013.

FIGURE 3: Effect on volume of Portuguese exports

This Figure presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time. We use annual data from 2011 to 2018. We present estimates for the average treatment effect over time. We include destination and firm-product-year fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. Errors are clustered by firm. We present 95 percent confidence intervals.



The presence of pre-trends poses an econometric challenge. Comparing the average evolution of exports to high-risk destinations before and after 2013 yields a coefficient with a positive bias, implying that our estimate represents a lower bound (in absolute value) of Basel III's effects on exports. However, pre-trends also introduce the possibility of alternative explanations for our findings.

As shown in Figure 2, the share of Portuguese exports directed to high-risk destinations steadily increased between 2000 and 2013. This trend accelerated after 2011 as Portuguese firms responded to the Eurozone crisis by expanding into non-EU markets. These forces are not fully captured by our empirical model. Specifically, the increase in exports to high-risk destinations triggered by the Eurozone crisis is not accounted for by destination controls, as it was primarily a supply-side response rather than a demand-driven shift. Similarly, firm-product-year fixed effects do not fully absorb this trend, as it varies across destinations, reflecting exporters' diversification beyond the EU.

We therefore attribute the observed pre-trends to two factors: a long-run trend of increasing exports to high-risk destinations and structural adjustments induced by the Eu-

rozone crisis. Both forces contribute to a positive bias in our estimates, suggesting that the true impact of higher credit costs on Portuguese exports may exceed 8 percent.

An alternative explanation for our findings is that demand for Portuguese exports from high-risk destinations declined after 2013.¹⁸ While this explanation is consistent with the model we presented, it would require that the decline in demand is specific to Portuguese exports rather than a general decline in demand from high-risk destinations.

In Appendix A, we show that after 2013, exports from EU countries to high-risk destinations declined. However, this pattern is not observed for non-EU countries—most of which did not implement Basel III. In 2013, non-EU countries sent 41 percent of their total exports to high-risk destinations; by 2018, this share remained virtually unchanged at 42 percent.¹⁹

Thus, for the demand-based explanation to hold, demand for Portuguese exports from high-risk destinations would need to have declined after 2013, while demand for the same products from non-EU exporters remained stable. However, this explanation is inconsistent with our findings in Section 5. A decrease in demand for Portuguese exports would reduce Portuguese firms' production and, consequently, their demand for credit. Lower credit demand, in turn, would be expected to reduce interest rates. Instead, we find evidence of rising interest rates, which aligns with a contraction in credit supply due to Basel III rather than a decline in export demand.

If we use the variation in exports from non-EU countries to high-risk destinations as a counterfactual for changes in demand, we conclude that demand for Portuguese exports remained stable between 2013 and 2018. Under this assumption, the coefficients in Figure 3 can be interpreted as the average treatment effects of Basel III.

If, instead, demand for Portuguese exports increased after 2013, then the coefficients in Figure 3 represent upper bounds of the true average treatment effect. To account for this possibility, we apply the method developed by Rambachan and Roth (2022) in Appendix A to compute confidence intervals under different assumptions about pre-trends. Even in this more conservative framework, we find persistent, statistically significant, and negative effects on exports following the introduction of Basel III.

¹⁸Another potential factor could be an increase in tariffs imposed by high-risk destinations on goods from Portugal or, more broadly, from the EU. However, as shown in Figure A.4, this is not the case. In fact, after 2013, tariffs on EU-origin goods in high-risk countries decreased, which would have mitigated the negative effects on exports that we identify.

¹⁹Table D.I in Appendix D further confirms that exports from non-EU countries to high-risk destinations did not decline across all products. In fact, there is some evidence of an increase in exports to high-risk destinations after 2013.

Heterogeneity across countries. We further estimate equation (2) by decomposing high-risk countries into three groups based on their OECD sovereign rating in 2013: (1) countries with a sovereign rating of 2 (e.g., China), (2) countries with a sovereign rating of 3 (e.g., Brazil), and (3) countries with a sovereign rating between 4 and 7 (e.g., Angola).²⁰ We use heterogeneity in sovereign risk within high-risk countries as a proxy for the likelihood that a bank is classified as high-risk. For instance, while some banks in China may be considered low-risk, most Angolan banks are likely classified as high-risk.²¹

This decomposition is informative because the probability that the foreign bank involved in a trade finance loan is high-risk (and therefore subject to a risk weight of 0.5) increases with the sovereign risk of the country (BIS, 2015). Moreover, this classification aligns with the EU’s methodology for assigning risk weights to non-trade finance claims on foreign banks, making it a familiar framework for financial institutions. The results of this decomposition are presented in Table III.

Basel III should lead to larger cost increases for banks financing exports to higher-risk destinations. Consequently, our mechanism predicts that countries with higher risk weights should experience a more pronounced decline in exports. Across all specifications, we find that the decline in exports is more severe for high-risk destinations with higher sovereign risk. In our preferred specification (column 4), exports to medium-risk countries (e.g., China) remain unchanged relative to exports to low-risk destinations. However, for countries with weaker sovereign ratings, we find a monotonic decline in exports as sovereign risk increases.

4.2 Product Mix

Production times vary across products. The longer the interval between the hiring of production inputs and the completion of output, the greater the firm’s need for working capital to cover labor and other costs. For Portuguese firms, bank credit is the primary source of working capital. Consequently, products with longer production cycles are more reliant on bank financing. As credit costs rise, the marginal cost increase should be more pronounced for credit-intensive products compared to those requiring little or

²⁰The OECD assigns sovereign ratings from 0 to 7, where 0 represents the safest countries and 7 the riskiest. All OECD member countries have a rating of 0 or 1 and are classified as low-risk. These ratings remain unchanged over our sample period.

²¹Our empirical analysis implicitly assumes that all treated countries experience the same intensity of treatment. However, the increase in risk weights is not uniform across high-risk countries. The presence of varying treatment intensities may bias estimates obtained from a two-way fixed effects estimator, as highlighted by De Chaisemartin and d’Haultfoeuille (2022) and Goldsmith-Pinkham et al. (2022). In Table A.III, we show that our estimates are robust to this concern.

no external financing.

Measure of dependence on credit. To compare products based on their reliance on working capital loans, we require a measure of credit dependence at the product level. However, no readily available dataset provides this information. We therefore follow the approach of [Rajan and Zingales \(1998\)](#) and [Chor and Manova \(2012\)](#), who use Compustat data to construct industry-level credit dependence measures for U.S. firms.²²

Our measure of credit dependence is the cash conversion cycle (CCC), which quantifies the time (in days) required for a firm to convert its investments in inventory and other resources into cash from sales. This is a particularly relevant measure for exporters, as it captures how long net input expenditures remain tied up before generating revenue. The CCC accounts for the time required to sell inventory, collect receivables, and settle payables. Formally, the CCC for firm i at time t is computed as:

$$\text{CCC}_{i,t} = \left(\frac{\text{Avg. Inventory}_{i,t}}{\text{COGS}_{i,t}} + \frac{\text{Avg. Accounts Receivable}_{i,t}}{\text{Sales}_{i,t}} \right) \times 365 \text{ days} \\ - \frac{\text{Avg. Accounts Payable}_{i,t}}{\text{COGS}_{i,t}} \times 365 \text{ days},$$

where the CCC depends on the ratio of average inventory to the cost of goods sold (COGS), the ratio of average accounts receivable to sales, and the ratio of average accounts payable to COGS. The numerators are computed as the average of beginning and ending balances for inventory, accounts receivable, and accounts payable. A high CCC indicates a longer lag between production investment and cash inflows, implying greater reliance on external credit. Conversely, a low CCC suggests a shorter cash flow cycle and reduced credit dependence.²³

We compute firm-level credit dependence in 2013 and aggregate it at the 3-digit NAICS industry level using a weighted average, with firm sales as weights. We then match these industry-level measures to 4-digit product codes using the methodology developed by

²²The literature frequently relies on U.S.-based measures of financial dependence, which offer distinct advantages. Our goal is to capture technological dependence on credit, independent of firms' reliance on bank credit or the financial development of a particular country. Since Compustat primarily includes large firms, we can be confident that our measure reflects fundamental technological needs rather than financial constraints. Moreover, given that the U.S. has one of the most developed financial systems globally, we interpret this estimate as a benchmark for credit dependence arising from *technological* constraints.

²³In our classification, manufacturing products tend to have higher CCC values. For example, cars have a CCC of 130 days, and engines 121 days. Similarly, pharmaceuticals and fertilizers exhibit high CCCs, at 124 and 117 days, respectively. In contrast, agricultural products typically have lower CCC values: coffee (44 days), meat products (42 days), and flour products (42 days).

Pierce and Schott (2009).²⁴

Results. We categorize products into two groups based on their cash conversion cycle (CCC): (1) high-dependence products, with a CCC above the median, and (2) low-dependence products, with a CCC below the median.²⁵

To assess the differential impact of Basel III on these two product groups, we estimate an augmented version of equation (2) that introduces a third difference across credit-dependence categories. The specification includes firm-product-year and destination fixed effects, as well as a vector of time-varying destination controls. Standard errors are clustered at the firm level. In this analysis, low-dependence products serve as the reference group. The results, based on data from 2011 to 2018, are presented in Table IV.

Across all specifications, we find that the decline in exports to high-risk destinations, relative to low-risk destinations, is more pronounced for products with a high dependence on bank credit. In our preferred specification (column 4), nearly all of the export decline to high-risk destinations is driven by a reduction in exports of credit-intensive products. This finding validates our proposed mechanism, which operates through the working capital channel, and highlights how firms respond to credit cost increases by adjusting their product mix. Specifically, firms shift production toward goods with lower working capital requirements to mitigate the impact of higher credit costs.

Our analysis reveals substantial heterogeneity in treatment effects. This heterogeneity poses challenges for standard estimation methods. As shown by De Chaisemartin and d’Haultfoeuille (2020), the average treatment effect estimated with a two-way fixed effects estimator is a weighted average of individual treatment effects, but the weights are not necessarily all positive, even though they sum to one. Consequently, it is possible for all individual treatment effects to be positive while the estimated average treatment effect appears negative. To address this issue, we employ the alternative estimator developed by De Chaisemartin and d’Haultfoeuille (2020), which is robust to heterogeneous treatment effects. The results, presented in Table A.II, are largely consistent with those obtained using the two-way fixed effects estimator, confirming the robustness of our findings.

²⁴We choose 3-digit NAICS industries to balance broad industry coverage with sufficient firm representation per industry. When multiple industries map to the same product, we aggregate industry-level measures using total industry sales as weights.

²⁵In Appendix A, we present the distribution of the CCC and show that high- and low-credit dependence products exhibit similar trade elasticities and product uses (e.g., intermediate, capital, or consumption goods). Additionally, both groups account for approximately 50 percent of total Portuguese exports and half of exports to high-risk countries.

4.3 Extensive Margin

We now turn to the extensive margin. The introduction of Basel III raises the cost of credit for exporters, leading to an increase in their marginal costs. In a firm entry and exit framework such as Melitz (2003), an increase in marginal costs should reduce net entry, even if entry costs remain unchanged.

Our data allows us to separately identify the effects on firm *entry* and *exit*. We do so at the market level, where we define a market as a product p and a destination d . We define an entrant as a firm that begins exporting product p to destination d in year t , having not exported that product to that destination in year $t - 1$. Conversely, an exiting exporter is a firm that was exporting product p to destination d in year $t - 1$ but ceases to do so in year t . We define entry and exit rates at the market level as follows:

$$\text{Entry rate}_{pdt} = \frac{\text{Number of entrants}_{pdt}}{\text{Number of exporters}_{pdt}}, \quad \text{Exit rate}_{pdt} = \frac{\text{Number of exiting exporters}_{pdt}}{\text{Number of exporters}_{pdt,t-1}}.$$

To compare the evolution of entry and exit rates in high-risk versus low-risk destinations, we estimate the following equations:

$$\text{Entry rate}_{pdt} = \lambda_{pt} + \mu_d + \beta X_{dt} + \gamma \times \mathbf{1}\{d \in \text{High-risk}\} \mathbf{1}\{t \geq 2014\} + u_{idt}, \quad (3)$$

$$\text{Exit rate}_{pdt} = \lambda_{pt} + \mu_d + \beta X_{dt} + \gamma \times \mathbf{1}\{d \in \text{High-risk}\} \mathbf{1}\{t \geq 2014\} + u_{idt}, \quad (4)$$

where we control for product-year and destination fixed effects, as well as time-varying destination characteristics. Standard errors are clustered at the destination level. The results of this analysis are presented in Table V.

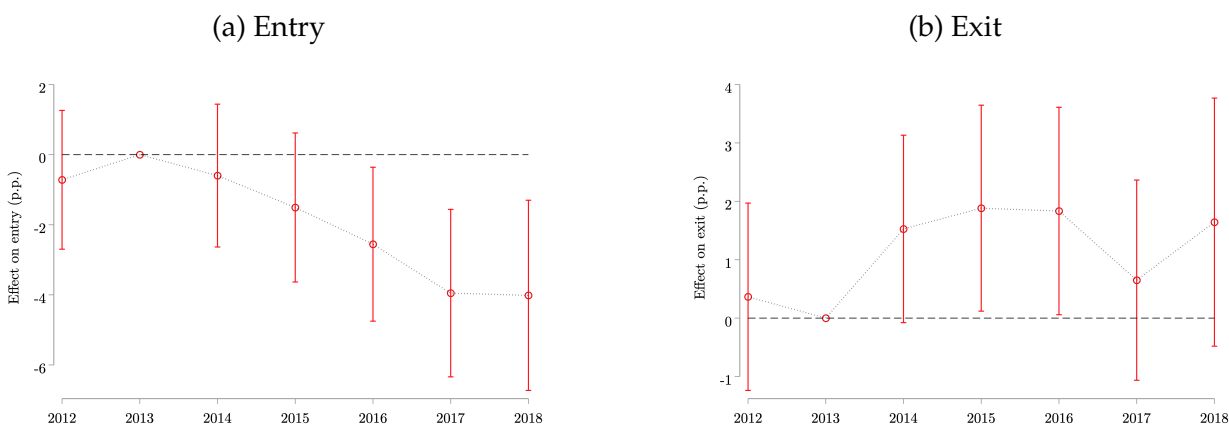
Results. We find that entry into high-risk destinations declines by 2.5 percentage points relative to entry into low-risk destinations, a result that is robust across various specifications. Given an average entry rate of 48 percent in 2013, this represents a 5 percent decline in entry. In contrast, exit from high-risk destinations increases by 1.3 percentage points, though this effect is not statistically significant.²⁶

²⁶In Table A.IV in Appendix A, we decompose the effects on entry and exit by high-risk country group. We find a modest increase in exit for countries with an OECD sovereign rating between 4 and 7, while the decline in entry is primarily driven by countries with either medium risk (OECD sovereign rating of 2) or very high risk (OECD sovereign ratings between 4 and 7). In Tables A.V and A.VI, we verify that our findings are not driven by heterogeneous treatment effects. In Tables A.VII and A.VIII, we further confirm that they are not influenced by the contamination bias discussed in De Chaisemartin and d'Haultfoeuille (2022) and Goldsmith-Pinkham et al. (2022). We find a decline in entry for countries with very high risk (OECD sovereign ratings between 4 and 7) and no significant effects on exit.

The effects on entry and exit into high-risk destinations over time are plotted in Figure 4.

FIGURE 4: Effects on entry and exit into destinations

This figure presents the results of estimating equations (3) and (4), where the dependent variable is the entry rate and the exit rate, respectively, and where we allow the average treatment effects to vary over time. The entry rate is the ratio of the number of entrants at time t to the number of firms at time t and the exit rate is the ratio of number of firms exiting between $t - 1$ and t to the number of firms in time $t - 1$. We use annual data from 2011 to 2018. We present estimates for the average treatment effect. We include product-year and destination fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. Errors are clustered by destination. We present 95 percent confidence intervals.



The decline in entry is not immediate, as shown in panel (a)—entry rates decrease steadily until 2017 before stabilizing. In contrast, the effects on exit, though small, are immediate: exit rates increase sharply in 2014 and then remain stable, as seen in panel (b). We attribute this difference in timing to how exporters learn about higher credit costs. Incumbent exporters, who regularly require credit, immediately experience the increase in interest rates on loans for exports to high-risk destinations and may exit accordingly. In contrast, potential entrants only become aware of higher borrowing costs when they seek financing for export activities, which introduces a delay in their response.

Interestingly, we find no significant effects on exit from high-risk destinations, indicating an asymmetric response on the extensive margin—entry declines, but exit remains unchanged. This asymmetry has been documented in response to temporary shocks, as in [Alessandria and Choi \(2007\)](#), who propose that the fixed costs of continuing to export are lower than the costs of entry. This explanation extends to permanent shocks: if the

cost of maintaining export operations is lower than the entry cost, firms will be more responsive in adjusting entry rather than exit. To see why, consider that a firm will only enter a destination if expected operating profits exceed the continuation cost plus a share of the total entry cost amortized over the firm’s lifetime. Conversely, a firm will only exit if its operating profits fall below the continuation cost. Since the threshold for entry is higher than that for exit, a decline in profitability leads to a stronger response in entry than in exit.

Our finding that credit shocks affect trade through the extensive margin is novel. [Paravisini et al. \(2015\)](#), in their study of Peruvian exporters, find no effects on entry or exit following a credit shock—an outcome inconsistent with firm heterogeneity models such as [Melitz \(2003\)](#). Two factors may explain why we detect entry effects while previous studies do not. First, we analyze a permanent credit shock, whereas most prior work, including [Paravisini et al. \(2015\)](#), focuses on temporary shocks. Second, our analysis spans six years after the shock, while [Paravisini et al. \(2015\)](#) consider only three years, potentially missing longer-run adjustments.

5 Effect on credit conditions

Thus far, we have focused on the impact of Basel III on exports. However, Basel III affects firms’ marginal costs, which in turn influence overall sales. To confirm that the regulatory shock led to an increase in marginal costs, we analyze the evolution of interest rates obtained by exporters.

Ideally, we would compare the interest rates on trade finance loans for high-risk and low-risk destinations within the same firm and product. However, this direct comparison is not feasible, as we cannot perfectly identify trade finance loans, nor can we allocate loans to specific export destinations within a firm. To address this limitation, we follow an approach similar to the one in [Khwaja and Mian \(2008\)](#) and compare firms based on their exposure to high-risk destinations. Firms with a greater share of exports to high-risk destinations are more likely to use loans for financing exports to those markets. This approach relies on the assumption that the probability of a loan being allocated to exports does not systematically vary with the firm’s share of exports to high-risk destinations.

For this analysis, we use a dataset at the firm-year-bank-loan level, where we observe loan k obtained by firm i from bank b in year t . We restrict the sample to exporters. Additionally, we include only new loans with maturities under 180 days. These filters ensure that we focus on loans that are either trade finance or sufficiently similar to serve as a valid control group. The credit registry data only starts in 2013 and so our sample starts

in 2013 and ends in 2018.

5.1 Effect on Loan Pricing

We start by looking at interest rates and loan amounts on all loans obtained by exporters. We classify firms into two groups based on their share of exports to high-risk destinations: firms above the cross-sectional median are categorized as *high-exposure*, while those below the median are classified as *low-exposure*.²⁷ We estimate the following regression:

$$Y_{ikbt} = \alpha_i + \lambda_{bt} + \gamma \times \mathbf{1}\{i \in \text{High-exposure}\} \cdot \mathbf{1}\{t \geq 2014\} + \beta W_{ikbt} + u_{ikbt}. \quad (5)$$

The dependent variable is either the interest rate or the logarithm of the loan amount. On the right-hand side, we control for firm and bank-year fixed effects, as well as loan characteristics, including loan maturity (in days) and an indicator variable that equals one if the loan is collateralized. Additionally, we include a vector of firm-level controls: log total sales, the sales-to-assets ratio, the leverage ratio, the EBITDA-to-assets ratio, the growth rate of total sales, labor productivity, the ratio of current-to-total liabilities, the ratio of current-to-total assets, and the firm's age squared. The parameter of interest, γ , estimates the average treatment effect. Standard errors are clustered at the firm level.

Discussion of Assumptions. The inclusion of firm fixed effects and firm-level controls in equation (5) ensures that identification of γ relies solely on variation driven by firms' export portfolios. Since most Portuguese exporters borrow from a single bank, firm fixed effects also capture the duration of the bank-firm relationship. Additionally, we include bank-year fixed effects to absorb shocks at the bank level. Given that the Portuguese banking sector is highly concentrated and banks are not well-capitalized, the implementation of Basel III likely affected bank lending beyond its direct impact on trade finance.

The main identification challenge arises from the fact that we cannot directly observe trade finance loans. As the cost of obtaining trade finance increases, exporters may attempt to substitute these loans with other forms of credit. If alternative loans carry lower interest rates, this substitution would introduce a negative bias in our estimates.²⁸

²⁷We exclude firms that export exclusively to low-risk or exclusively to high-risk destinations. This ensures consistency with our empirical analysis in Section 4, where firm-product-year fixed effects and the interaction between year fixed effects and the high-risk destination indicator imply that firms exporting to only one type of destination are not included in the regression.

²⁸In Table B.I, we show that the total number of loans obtained by a firm in a given year remains unchanged, suggesting limited substitution across loan types.

Results. The results of this estimation are presented in Table VI. We find that firms primarily exporting to high-risk destinations obtain smaller loans after the implementation of Basel III compared to firms that mainly export to low-risk destinations. This result is robust across all specifications. In our preferred specification (column 3), firms exporting predominantly to high-risk destinations receive loans that are, on average, 7 percent smaller than those obtained by firms exporting to low-risk destinations. This effect is of the same order of magnitude as the decline in exports to high-risk destinations (approximately 8 percent), suggesting an estimated elasticity of exports to loan amounts close to one.²⁹

In contrast, interest rates are far less responsive. In column (3), we estimate that firms primarily exporting to high-risk destinations face an interest rate that is 13 basis points higher than the rate faced by firms exporting to low-risk destinations. Given that the average interest rate in 2013 was 7.3 percent, this corresponds to a 1.8 percent increase.

This asymmetric response between loan amounts and interest rates can be partly explained by firms adjusting loan sizes in response to higher borrowing costs. Basel III increases the cost of extending trade finance to firms exporting to high-risk destinations. Consequently, when seeking loans, firms attempt to mitigate the cost increase by moving along the demand curve and opting for smaller loans. Additionally, credit rationing and borrower selection effects may contribute to the muted response in interest rates.³⁰

Our findings in Table VI suggest that the observed decline in exports is driven by a contraction in credit supply rather than a reduction in foreign demand for Portuguese goods. If the decline in exports were demand-driven, Portuguese exporters would scale back production, reducing their need for working capital loans.³¹ This decline in credit demand would, in turn, put downward pressure on interest rates. However, we observe the opposite pattern—interest rates increase—indicating that the primary driver of the export decline is a tightening of credit supply rather than weaker external demand.

²⁹One potential concern is that firms may substitute bank financing with trade credit from customers or suppliers. Indeed, we find that receivables decline during this period, indicating that firms increasingly rely on their customers for financing. This finding is also consistent with the observation that receivables tend to be higher for exports to high-risk destinations, so as exports to these markets decline, so do receivables. Additionally, we observe a reduction in inventories, which we present in Table A.IX.

³⁰To examine dynamic effects, we allow the average treatment effect to vary over time. The results, presented in Figure B.1, show that loan amounts decline immediately and continue decreasing until 2017. In 2014 and 2015, firms primarily exporting to high-risk destinations experience interest rate increases exceeding 20 basis points relative to firms exporting to low-risk destinations.

³¹In Supplementary Appendix F, we also show that exporters cut wages and the number of workers, which is consistent with a decrease in labor demand.

5.2 Effect on Credit Rationing

Our estimates in Table VI are likely biased due to adjustments along the extensive margin. Exporters that exit or cease entering high-risk destinations will naturally seek fewer loans. Moreover, these firms tend to be the least productive, meaning that their exit increases the average productivity of firms exporting to high-risk destinations. Since banks typically charge lower interest rates to more productive firms, this selection effect should lead to a decline in observed interest rates, biasing our estimates downward.

Beyond selection, credit markets also feature rationing. When credit rationing is present, the equilibrium interest rate is not fully informative, as some firms would be willing to borrow at higher rates but are unable to do so due to lending constraints. Both selection and rationing imply that firms primarily exporting to high-risk destinations should receive fewer loans.

To test this, we use our credit registry data to estimate the following equation:

$$\text{Receives loan}_{it} = \alpha_i + \lambda_t + \gamma \times \mathbf{1}\{i \in \text{High-exposure}\} \cdot \mathbf{1}\{t \geq 2014\} + \beta W_{it} + u_{it}, \quad (6)$$

where the dependent variable is an indicator equal to 1 if firm i receives at least one bank loan in year t , and 0 otherwise. The specification includes firm and year fixed effects, along with a vector of firm-level controls. Standard errors are clustered at the firm level. The results of this estimation are presented in Table VII.

We find that firms primarily exporting to high-risk destinations experience a 5.9 percentage point decline in the probability of obtaining a loan compared to firms that mainly export to low-risk destinations. This effect is robust across various specifications. Given the pre-policy average probability of obtaining a loan, this decline represents a 7 percent drop.

Thus, firms with high exposure to high-risk destinations not only receive fewer loans but, conditional on obtaining a loan, face higher interest rates and smaller loan amounts. Moreover, the reduction in loan probability suggests that the firms that continue to secure financing are those with the highest credit quality and lowest risk. This selection effect further dampens the estimated impact of Basel III on interest rates, as lower-risk firms are likely to face more favorable lending terms.³²

³²Allowing the average treatment effect to vary over time, we present the results in Figure B.2. In response to Basel III, the probability that a firm primarily exporting to high-risk destinations obtains a loan declines immediately by nearly 6 percentage points. This effect is persistent, as the decline in loan probability continues through 2018. The permanence of this effect is consistent with a lasting increase in the cost of bank credit for firms exporting to high-risk destinations.

6 Model

In our empirical analysis in Section 5, we find that the impact of Basel III on interest rates is limited. We attribute this finding to credit rationing—if market interest rates do not clear the credit market, they are not a useful statistic for assessing the impact of Basel III. Under credit rationing, banks may be unwilling to lend to certain borrowers even at interest rates above the market rate. As a result, the true cost of Basel III is captured not by observed interest rates, but by the *shadow cost* of credit.

To estimate this shadow cost, we require additional structure—specifically, a model that links interest rates on export financing to trade flows. By combining our empirical estimates with aggregate trade data for the European Union, we can invert this relationship to infer the shock to interest rates that rationalizes the observed trade patterns. We interpret this inferred shock as the shadow cost of Basel III.

Additionally, the model allows us to quantify the effects of Basel III on welfare through its impact on international trade. However, it is important to clarify that our model does not capture the overall welfare effects of Basel III. In our framework, the increase in interest rates is treated as exogenous. In reality, bank regulation is endogenous and aims to correct distortions in lending behavior. The regulatory intent behind Basel III was to address excessive risk-taking by banks, particularly in lending to exporters selling to high-risk destinations. If banks were previously underpricing risk, Basel III could improve welfare in EU countries by mitigating financial distortions. The same logic does not apply to high-risk countries. For these countries, Basel III solely increases the cost of imports, which unambiguously reduces welfare. Therefore, we interpret the welfare changes in our model as the welfare costs of Basel III rather than its full welfare implications.

6.1 Model setup

We consider a multi-country, multi-sector Ricardian model of international trade following [Caliendo and Parro \(2015\)](#). The model is static, with N countries indexed by either n or i and, and J sectors indexed by j or k . Throughout the model, subscripts denote countries, while superscripts denote sectors.

Trade arises because firms seek to minimize input costs, but households do not consume foreign goods. Each sector contains two types of goods: composite goods and intermediate goods. Composite goods are non-tradable and produced using intermediates from the same sector. Producers of composite goods may import intermediates. Intermediate goods, in contrast, use domestic composite goods from all sectors and labor as

inputs and may be tradable.

We provide a detailed description of the model structure in Appendix C. Here, we focus on the key modifications relative to [Caliendo and Parro \(2015\)](#) and examine how changes in interest rates affect welfare.

Trade costs and prices Trade is costly due to three types of frictions: (1) iceberg costs, (2) tariffs, and (3) financial frictions. Iceberg costs are standard: shipping one unit of good j from country n to country i requires sending $d_{ni}^j \geq 1$ units, where $d_{nn}^j = 1$. Country n imposes an ad-valorem tariff $\tau_{in}^j \geq 0$ on goods imported from country i , applied to unit prices, with $\tau_{nn}^j = 0$. Firms in sector j in country n that export to country i must finance production costs upfront by borrowing from a bank at an interest rate $r_{ni}^j \geq 0$, which they take as given. Firms selling domestically do not face this constraint and so $r_{nn}^j = 0$. These three trade costs can be combined into a single factor κ_{ni}^j that scales unit costs:

$$\kappa_{ni}^j = d_{ni}^j \times (1 + \tau_{ni}^j) \times (1 + r_{ni}^j).$$

Iceberg costs represent a deadweight loss. Tariff revenues, in contrast, are rebated to the representative household as a lump-sum transfer. We assume that financial frictions behave like iceberg costs, meaning that interest payments made by firms are not rebated to households but instead represent a loss of resources. In practical terms, this means that the total value of interest payments enters the aggregate resource constraint but not the household's budget constraint, whereas tariff revenues are redistributed lump-sum.

Welfare In this model the natural measure of welfare is real household income, or real GDP. The following Proposition shows that we can decompose changes in welfare into three terms: a terms-of-trade effect, a volume-of-trade effect, and a trade costs effect.

Proposition 1 *The change in welfare in country n arising from a change in interest rates can be written as*

$$\begin{aligned} d \log W_n = & \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(E_{ni}^j d \log c_n^j - M_{in}^j d \log c_i^j \right) \\ & + \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j \left(d \log M_{in}^j - d \log c_i^j \right) \\ & - \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(1 + \tau_{in}^j \right) M_{in}^j dr_{in}^j. \end{aligned} \tag{7}$$

where W_n is welfare in country n , I_n is total household income (wage income and tariff revenue), E_{ni}^j denotes exports of product j by country n to country i , c_n^j is the marginal cost of producing good j in country n net of trade costs, and M_{in}^j denotes imports of good j from country i to country n .

Proof. In Supplemental Appendix E. ■

The first term captures a terms-of-trade effect, reflecting the change in the relative price of country n 's exports compared to its imports. This effect depends only on marginal costs, not on interest rates or tariffs, because revenue generated by these frictions is not retained by firms. An improvement in a country's terms of trade increases welfare, as it allows a given quantity of exports to command a larger quantity of imports in the world market.

The second term measures the welfare gains from changes in trade volume, specifically the quantity of imports. An increase in imports raises welfare through two channels: (1) higher trade volumes generate more tariff revenue for the country, and (2) greater access to imported intermediates lowers production costs, thereby increasing productivity.

The third term represents the welfare losses due to rising interest rates. If interest rates increase for exports from country i to country n , then country n will face higher import prices, reducing productivity and lowering welfare. For a high-risk country, this represents the direct effect of an increase in interest rates. In contrast, for country i , the effect operates through general equilibrium adjustments: as demand for its exports declines, its marginal costs fall. Thus, we can interpret this third term in equation (7) as the first-round impact of rising interest rates while holding trade flows and prices constant.

Consider a high-risk country. As interest rates increase, high-risk countries experience welfare losses due to higher trade costs. The magnitude of this effect depends on the scale of EU exports to the high-risk country. Higher import prices force a reduction in imported quantities, leading to a negative volume-of-trade effect as tariff revenues decline. Additionally, as firms must use more expensive imported intermediates, marginal costs rise. If this increase in marginal costs exceeds the rise in import prices, the high-risk country may experience a positive terms-of-trade effect.

Consider now a EU country. Since the interest rates imposed on EU countries' imports remain unchanged, the cost-of-trade effect is zero—keeping trade flows and prices constant, welfare in EU countries does not change because the additional revenue from interest payments is not distributed to households. However, rising interest rates reduce global demand for EU exports, leading to lower EU export volumes. By the balanced trade equation, this also necessitates a decline in EU imports, generating a negative volume-of-trade effect. Furthermore, as external demand for EU exports decreases,

domestic demand for production inputs—including labor—also declines. Given that labor supply is perfectly inelastic, this reduction in demand leads to higher nominal wages, which in turn raises domestic production costs. This increase in costs results in a negative terms-of-trade effect, assuming import prices remain unchanged.

Discussion In our model, we assume that revenue generated through interest rates represents a pure resource loss. An alternative assumption would be to treat this revenue as redistributed to households, similar to how we model tariff revenues. This alternative would introduce an additional channel through which higher interest rates affect welfare.

Under our baseline assumption, an increase in interest rates reduces welfare primarily by decreasing global demand for EU exports. As a result, domestic demand for production inputs, including labor, declines, leading to lower wages. The reduction in wages decreases household income, further amplifying the welfare loss.

If, instead, interest rate revenue were redistributed to households, there would be an offsetting *income effect*. Higher interest rates would generate additional revenue, increasing household income. This, in turn, would boost demand, stimulate factor markets, and raise wages, partially mitigating the welfare loss from reduced exports.

However, our model does not account for the frictions that originally motivated Basel III. The regulation was introduced because policymakers believed that banks were mispricing financial products, particularly trade finance. If banks systematically underestimated the risk associated with lending to high-risk destinations, they would have charged interest rates that were too low. In our framework, interest rate changes are exogenous, and there is no underlying financial risk. Thus, redistributing interest rate revenue to households would not provide meaningful insight into the true impact of Basel III.

Moreover, the purpose of Basel III was not to increase bank profits or shareholder dividends, but rather to correct financial mispricing. Our analysis focuses on the costs of Basel III, not its potential benefits. Including interest rate revenue as household income would lead to an underestimation of these costs without yielding useful information about the regulation’s intended benefits.

6.2 Impact of Basel III

Following [Caliendo and Parro \(2015\)](#), we calibrate the model using changes rather than levels. Since aggregate trade deficits are treated as exogenous, we begin by eliminating them and constructing a no-deficit world economy as our base year. Importantly, this approach does not require sectoral trade balances to be zero, nor does it directly determine

their values, as they remain endogenous within the model.

Identification of the shock. We assume the following specification for the increase in interest rates:

$$\Delta r_{in}^j = \begin{cases} \zeta & \text{if } i \in \text{E.U. and } n \in \text{High-risk and } j \in \text{High-credit,} \\ \phi \times \zeta & \text{if } i \in \text{E.U. and } n \in \text{High-risk and } j \in \text{Low-credit,} \\ 0 & \text{if otherwise.} \end{cases}$$

For high-credit sectors, we assume that interest rates increase by $\zeta \geq 0$ for exports from the EU to high-risk countries. For low-credit sectors, we assume that interest rates increase by $\phi \times \zeta$ where $\phi \in [0, 1]$. This specification can be micro-founded in a framework where firms finance only a fraction ϕ of their marginal costs through external borrowing, using internal funds for the remainder.

We classify tradable sectors as either high-credit dependent or low-credit dependent.³³ The two parameters we estimate in our empirical exercise for 2018 allow us to calibrate these values. However, in our framework, we cannot separately identify the exposure of high-credit and low-credit products. Consequently, when identifying ζ , we can only estimate the ratio $\phi^{\text{low}} / \phi^{\text{high}}$. By imposing the normalization $\phi^{\text{high}} = 1$ we can identify ϕ^{low} .

We use the two parameters $\{\zeta, \phi\}$ to match the average treatment effect on exports to high-risk destinations observed in the data. To compute this effect, we conduct an analysis similar to that in Section 4, but using aggregate trade data for EU countries. Specifically, we estimate average treatment effects for high- and low-credit products by leveraging within-exporter-product variation, before and after the introduction of Basel III. Further details on this estimation procedure are provided in Appendix C.

It is important to note that this approach does not involve matching the unconditional growth rate of high-credit exports from EU countries to high-risk destinations. Instead, for each parameter vector, we simulate a path for exports and estimate a triple-difference regression, following the methodology in Section 4. The results of our estimation exercise are presented in Table VIII.

Our model estimates an increase in interest rates of 1.75 percentage points for exports of high-credit products from EU countries to high-risk destinations. Additionally, we match the effect on low-credit products by calibrating the exposure parameter ϕ to 0.4.

³³Sectors are classified using the cash conversion cycle measure. A full list of sector classifications is provided in Appendix C.

This aligns with our empirical findings, where the impact on high-credit products is approximately 60 percent larger than that on low-credit products. This result suggests that for low-credit products, firms only need to finance 40 percent of their marginal costs in advance.

The interest rate changes documented in Table VIII are significantly larger than those estimated in Section 5. In our empirical analysis using credit registry data, we found that firms primarily exporting to high-risk destinations faced an increase of only 13 basis points in interest rates. By contrast, our model suggests an increase an order of magnitude higher.

This discrepancy arises because our calibration captures the shadow cost of Basel III rather than the observed market interest rate. In the presence of credit rationing, the market interest rate does not fully reflect the true costs imposed by Basel III. Additionally, in our model, firms cannot exit or opt out of exporting to high-risk destinations, which prevents the dampening effect observed in the empirical estimation of interest rate changes. Finally, firms in the model are not allowed to substitute bank loans with alternative sources of external credit, ensuring that we fully capture the impact of Basel III on borrowing costs. As a result, our estimation strategy provides a more comprehensive assessment of the costs imposed by Basel III through its effects on international trade.

Our model closely replicates the growth rate of exports to high-risk destinations. Importantly, this is not a calibrated moment, as our exercise seeks to match the conditional relative growth rate of exports to high-risk countries, while the observed growth rate is unconditional and may be influenced by additional shocks not included in the model. The close alignment between these two growth rates lends credibility to our mechanism.

However, we are unable to match the growth rate of exports to low-risk destinations. This discrepancy arises from our empirical strategy, which uses exports to low-risk countries as a control group. By construction, we fully absorb changes in this group using fixed effects, meaning that any shifts in demand, unobserved productivity changes, or trade cost adjustments for low-risk destinations are not accounted for in our model. Consequently, we also underestimate the total growth of exports, as we miss part of the expansion in exports to low-risk destinations.

Finally, while our model successfully captures the change in the share of exports to high-risk countries, it underestimates the magnitude of the decline. This mismatch stems from our underestimation of the growth rate of exports to low-risk destinations.

Effect on welfare. Using the decomposition outlined in equation (7), we compute the change in real GDP for all countries in our sample, along with the contributions from

changes in terms of trade, trade volume, and credit costs. The results are presented in Table IX.

EU countries experience a welfare loss of 0.09 percent due to the increase in interest rates. While this may appear small, Arkolakis et al. (2012) note that the overall impact of trade costs on welfare tends to be limited. However, within the context of typical trade-related welfare effects, this loss is substantial. For comparison, Caliendo and Parro (2015), using a similar model, estimate that the total gains from NAFTA for the U.S. were 0.08 percent. This suggests that the costs of Basel III for the EU are comparable in magnitude to the trade benefits the U.S. derived from NAFTA.

The cost-of-trade effect is zero, as interest rates on exports from non-EU countries remain unchanged. The volume-of-trade effect is negative but minimal, indicating that the EU did not substantially reduce imports in response to higher interest rates—total EU imports declined by only 1.18 percent.

The primary driver of the welfare loss is the terms-of-trade effect. As EU exports become less competitive due to higher financing costs, global demand declines. This reduction in demand leads to lower factor prices, including wages. In the case of labor, which is assumed to have perfectly inelastic supply, a decline in demand translates directly into lower nominal wages. The drop in factor prices reduces marginal costs, which in turn lowers export prices. This decline in export prices has a mixed impact: on one hand, it increases the competitiveness of EU exports, partially offsetting the negative volume-of-trade effect. On the other hand, it reduces the EU's ability to finance imports with the same level of exports, which negatively impacts welfare.

Interestingly, the price of imports declines, creating a positive effect on welfare. This occurs because the EU's primary trading partners are other EU countries. As a result, when a trading bloc collectively raises interest rates, it mitigates some of the potential welfare losses by reducing import prices within the bloc.

High-risk countries also experience welfare losses of 0.04 percent, driven entirely by the first-round effects of higher interest rates. Both the volume-of-trade effect and terms-of-trade effect are small, indicating that high-risk countries do not significantly reduce their import quantities and that their terms of trade remain stable or improve slightly. The improvement in terms of trade is largely driven by rising export prices, which result from higher marginal costs. These cost increases stem from the higher price of imported inputs, making production more expensive.

Notably, welfare losses are larger for EU countries than for high-risk countries. This outcome reflects the structure of global trade flows: while high-risk countries face higher financing costs, their imports are only marginally affected, and their terms of trade may

even improve. In contrast, EU countries suffer from declining global demand for their exports, lower factor prices, and reduced purchasing power in international markets.

7 Conclusion

In this paper, we study how exporters react to an increase in the cost of credit. We examine three channels: the intensive margin, the extensive margin, and product mix adjustments. We find that, in response to an increase in the cost of credit, exporters cut exports to high-risk destinations by 8 percent. Moreover, overall exports do not increase, which suggests exporters cannot offset the decrease in exports with an increase in exports to other locations. On the extensive margin, we find that entry into high-risk destinations declines. We also show that firms' product mix changes in response to this shock. Firms adjusted by shifting towards products with a lower dependence on working capital and bank credit. This finding underscores the importance of product characteristics, such as the cash conversion cycle, in determining a firm's overall exposure to credit shocks. We also find that firms respond to the increase in the cost of credit by reducing the size of the loans they obtain from banks or by obtaining fewer loans overall.

We also use a structural model, which we calibrate using our empirical results to identify the welfare consequences of this increase in the cost of credit. This model combines a Ricardian trade structure with financial frictions. Using the model, we find that the increase in the cost of trade leads to welfare losses for high-risk destinations as the price of imports increases. It also leads to welfare losses for EU countries due to a depreciation in terms of trade caused by an erosion of comparative advantage.

Our analysis focuses on the impact of Basel III on exporters. However, importers may also be reacting to the increase in costs. For example, do overall imports of firms in high-risk destinations fall? Does this decline vary across firms? Do these importers shift their banking relationships to banks that are perceived as low-risk? All of these questions are crucial to understanding the full impact of Basel III on trade, and we leave them for future research.

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TABLE III: Effects on volume of Portuguese exports - decomposition by sovereign risk of destination

This table presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time, and where compare the period between 2011 and 2013 with the period after 2014. We use annual data from 2011 to 2018. We present estimates for the average treatment effect decomposed across three groups of high-risk destinations: (1) medium risk countries with an OECD sovereign rating of 2, (2) high risk countries with an OECD sovereign rating of 3, and (3) very high risk countries with an OECD sovereign rating between 4 and 7. We include destination and firm-product-year fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also interact an indicator variable which takes the value of 1 if the destination country has Portuguese as an official language and zero otherwise with time fixed effects, and we interact an indicator variable which takes the value of 1 if the destination country was a former Portuguese colony after 1945 with time fixed effects. Errors are clustered by firm. ***, **, * denote significant at the 1, 5 and 10 percent levels respectively.

	(1)	(2)	(3)	(4)
Medium risk $\times$ Post 2014	-0.012 (0.052)	-0.025 (0.052)	-0.025 (0.051)	-0.032 (0.053)
High risk $\times$ Post 2014	-0.049 (0.031)	-0.088*** (0.032)	-0.086*** (0.032)	-0.062* (0.038)
Very high risk $\times$ Post 2014	-0.058 (0.036)	-0.119*** (0.036)	-0.116*** (0.036)	-0.096*** (0.037)
Destination FE	✓	✓	✓	✓
Firm $\times$ Product $\times$ Year FE	✓	✓	✓	✓
Common language $\times$ Year FE		✓		
Former colony $\times$ Year FE			✓	
Destination controls				✓
Observations	895,117	894,891	895,117	895,042

TABLE IV: Effects on volume of Portuguese exports - decomposition by product dependence on credit

This table presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time, and where compare the period between 2011 and 2013 with the period after 2014. We use annual data from 2011 to 2018. We present estimates for the average treatment effect for low-credit products, and the difference between average treatment effects for high-credit and low-credit products. We define a product as high-credit if its measure of credit dependence is above the median in 2013. We consider one measure of credit dependence: the cash conversion cycle (CCC). We include destination and firm-product-year fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. Errors are clustered by firm. ***, **, * denote significant at the 1, 5 and 10 percent levels respectively.

	(1)	(2)	(3)	(4)
High-risk $\times$ Post 2014	-0.008 (0.026)	-0.007 (0.027)	-0.063* (0.036)	-0.020 (0.037)
High-risk $\times$ High-credit $\times$ Post 2014	-0.054* (0.028)	-0.060** (0.026)	-0.069* (0.039)	-0.070* (0.038)
Destination FE	✓	✓	✓	✓
Year FE	✓	✓		
Firm FE	✓	✓		
Product FE	✓	✓		
Firm $\times$ Product $\times$ Year FE			✓	✓
Destination controls		✓		✓
Observations	1,651,763	1,640,139	898,667	888,563

TABLE V: Effects on entry and exit into destinations

This table presents the results of estimating equations (3) and (4), where the dependent variable is the entry rate and the exit rate, respectively. The entry rate is the ratio of the number of entrants at time t to the number of firms at time t and the exit rate is the ratio of number of firms exiting between $t - 1$ and t to the number of firms in time $t - 1$. We use annual data from 2011 to 2018. We present estimates for the average treatment effect. We include product-year and destination fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also present the average entry and exit rates for 2013. Errors are clustered by destination. *,** and *** denote significance at the 10, 5 and 1 percent levels, respectively.

	Entry			Exit		
	(1)	(2)	(3)	(4)	(5)	(6)
High-risk $\times$ Post 2014	-0.023** (0.011)	-0.020* (0.010)	-0.025** (0.011)	0.041*** (0.010)	0.035*** (0.010)	0.013 (0.009)
Mean in 2013	.484	.484	.484	.273	.273	.273
Destination FE	✓	✓	✓	✓	✓	✓
Year FE	✓			✓		
Product FE	✓			✓		
Product $\times$ Year FE		✓	✓		✓	✓
Destination controls			✓			✓
Observations	202,950	202,494	171,702	193,622	175,617	172,103

TABLE VI: Effect on interest rates and loan amounts

This table presents the results of estimating regression (5), where the dependent variable is the interest rate or the loan amount for loan k obtained by firm i from bank b in year t . We use individual loan data for exporters from 2013 to 2018, and consider only loans with maturities under 180 days. We present estimates for the average treatment effect. The time-varying loan controls include: the loan maturity and a dummy variable that takes the value of 1 if the loan is collateralized and 0 if otherwise. The time-varying firm controls include: log of total sales, the sales-to-asset ratio, the leverage ratio, the EBITDA-to-assets ratio, the growth rate of total sales, labor productivity, the ratio of current-to-total liabilities, the ratio of current-to-total assets, as well as the firm's age and its age squared. Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	Loan amounts			Interest rates		
	(1)	(2)	(3)	(4)	(5)	(6)
High exposure $\times$ Post 2014	-0.067** (0.026)	-0.063** (0.028)	-0.072** (0.026)	0.084 (0.064)	0.124* (0.066)	0.133** (0.062)
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓			✓		
Bank FE	✓			✓		
Bank $\times$ Year FE		✓	✓		✓	✓
Controls	✓		✓	✓		✓
Observations	793,984	795,077	793,981	793,984	795,077	793,981

TABLE VII: Effects on probability of obtaining a loan

This table presents the results of estimating regression (6), where the dependent variable is an indicator variable which takes the value of 1 if firm i receives at least one bank loan in year t and zero if otherwise. We use individual loan data for exporters from 2013 to 2018, and consider only loans with maturities under 180 days. We present estimates for the average treatment effect. The time-varying firm controls include: log of total sales, the sales-to-asset ratio, the leverage ratio, the EBITDA-to-assets ratio, the growth rate of total sales, labor productivity, the ratio of current-to-total liabilities, the ratio of current-to-total assets, as well as the firm's age and its age squared. Errors are clustered by firm. *,** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	(1)	(2)	(3)	(4)
High exposure $\times$ Post 2014	-0.063*** (0.008)	-0.064*** (0.008)	-0.057*** (0.008)	-0.059*** (0.008)
Mean in 2013	0.79	0.79	0.79	0.79
Firm FE		✓		✓
Year FE		✓		✓
Controls			✓	✓
Observations	66,954	66,954	61,444	61,156

TABLE VIII: Model calibration with heterogeneous effects

This Table presents the results of the calibration of our model with heterogeneous effects. We have two calibration targets: the average treatment effect on products with low credit dependence and the average treatment effect on products with high-credit dependence. We calibrate two parameters: the interest rate shock ζ and the exposure of low-credit products to interest rates ϕ . We also present four untargeted moments: the growth rate of total exports, the growth rate of exports to low-risk countries, the growth rate of exports to high-risk countries, and the change in the share of high-risk countries in total exports.

Panel A: Targeted Moments

Parameters	Value	Value	Data	Model
ζ	1.75 p.p.	Change in exports of high-credit products	-8.24%	-8.35%
ϕ	0.40	Change in exports of low-credit products	-5.31%	-5.39%

Panel B: Untargeted Moments

	Moment	Data	Model
	Growth in total exports (%)	4.72	-1.18
	Growth in exports to low-risk countries (%)	7.39	1.00
	Growth in exports to high-risk countries (%)	-3.86	-4.01
	Change in share of exports to high-risk countries (p.p.)	-1.94	-1.25

TABLE IX: Changes in welfare

This table presents the effects of the increase in interest rates on the welfare of all E.U. and high-risk countries in our sample. We decompose the change in welfare into three components: a terms-of-trade effect, a volume-of-trade effect, and a trade cost effect. We further decompose the terms-of-trade effect into the effect from the change in the price of exports and the effect from the change in the price of imports. All values represent the percentage change. We also present aggregated effects for all E.U. countries and for all high-risk countries by taking the simple average across all countries in the group.

	Terms of trade				Total
	Price of exports	Price of imports	Volume of trade	Credit costs	
E.U. countries	-0.82	0.73	-0.01	0.00	-0.09
Portugal	-0.81	0.75	0.00	0.00	-0.06
High-risk countries	0.31	-0.29	-0.01	-0.04	-0.04
Brazil	0.38	-0.36	-0.02	-0.04	-0.04
China	0.45	-0.43	-0.01	-0.04	-0.03
India	0.33	-0.32	-0.01	-0.04	-0.03

Online Appendix

A Appendix to Section 4

A.1 Addressing pretrends

FIGURE A.1: Effects on volume of exports under a linear trend

This Figure presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time. We use annual data from 2011 to 2018. We present estimates for the average treatment effect over time. We include destination and firm-product-year fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. Errors are clustered by firm. We present 95 percent confidence intervals for two specifications. In the first one, “No demand shocks”, we assume that $\delta_t = 0$ for $t > 2013$. In the second one, “Linear trend”, we assume a linear trend.

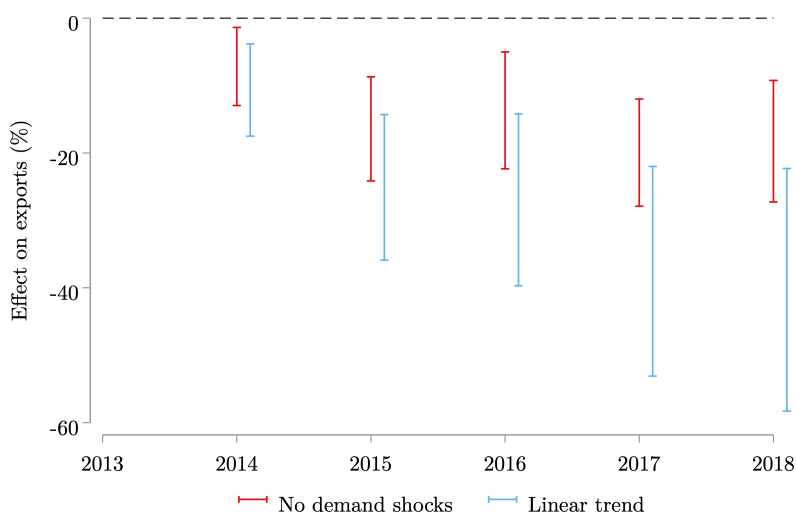
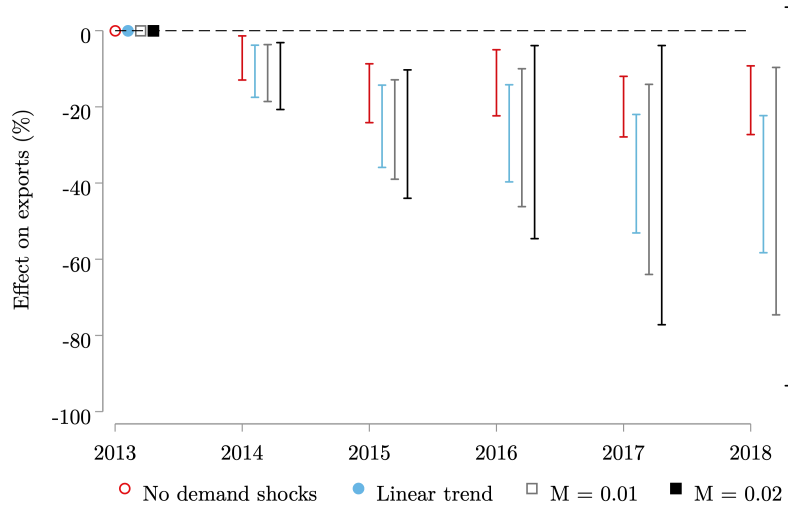


FIGURE A.2: Effects on volume of exports under smoothness restrictions

This Figure presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time. We use annual data from 2011 to 2018. We present estimates for the average treatment effect over time. We include destination and firm-product-year fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. Errors are clustered by firm. We present 95 percent confidence intervals for four specifications. In the first one, “No demand shocks”, we assume that $\delta_t = 0$ for $t > 2013$. In the second one, “Linear trend”, we assume a linear trend. In the third and fourth specifications we assume $M = 0.01$ and $M = 0.02$ respectively.



A.2 Additional results

FIGURE A.3: Distribution of Cash Conversion Cycle

This Figure presents the kernel estimate of the density function of the cash conversion cycle across products. We compute the cash conversion cycle for all U.S. firms in Compustat for 2013. The cash conversion cycle is defined as the sum of three components: (1) average inventory / cost of goods sold, (2) average accounts receivable / sales and (3) average accounts payable / cost of goods sold. We then multiply the result by 365. We then aggregate this measure at the industry level by taking the weighed average of the CCC, using the sales of the firm as the weight. We then match each industry to a product. We also show the median of the distribution in the dashed vertical line.

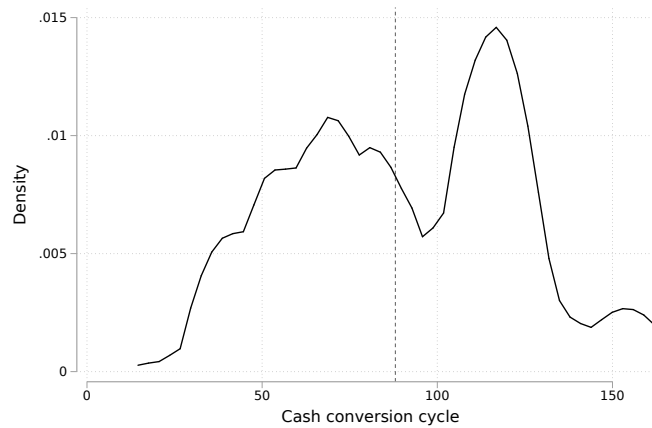


FIGURE A.4: Change in tariffs imposed by high-risk destinations

This figure presents the results of a two-way fixed effects regression of tariffs imposed by high-risk destinations. We use all tariffs at the importer-exporter-product-year level between 2011 and 2018 (where we aggregate products to the 4-digit level). We include only observations where the importer is a high-risk country. We regress the tariff on an importer-product-year fixed effect and on an exporter fixed effect. We present the coefficients associated with the interaction between a time fixed effect and an indicator variable which takes the value of one if the exporter is an E.U. country. We interpret each of these coefficients as the differential change in tariffs imposed on E.U. countries vs. other countries, relative to a base year (2013). If the coefficient is negative after 2013, it means that high-risk countries are lowering tariffs on goods imported from E.U. countries relative to other countries. We cluster the errors at the importer level and present 95 percent confidence intervals.



TABLE A.I: Effects on volume of exports

This table presents the results of estimating regression (2), where the dependent variable is the log of exports of firm i of product p to destination d at time, and where compare the period between 2011 and 2013 with the period after 2014. We use annual data from 2011 to 2018. We present estimates for the average treatment effect. We include destination and firm-product-year fixed effects, and we also present a specification with destination, time, product and firm fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also interact an indicator variable which takes the value of 1 if the destination country has Portuguese as an official language and zero otherwise with time fixed effects, and we interact an indicator variable which takes the value of 1 if the destination country was a former Portuguese colony after 1945 with time fixed effects. Errors are clustered by firm. ***, **, * denote significant at the 10, 5 and 1 percent levels respectively.

	(1)	(2)	(3)	(4)
High-risk $\times$ Post 2014	-0.050* (0.029)	-0.094*** (0.029)	-0.092*** (0.029)	-0.076** (0.033)
Share of exports going to high-risk (%)	22	22	22	22
Destination FE	✓	✓	✓	✓
Firm $\times$ Product $\times$ Year FE	✓	✓	✓	✓
Common language $\times$ Year FE		✓		
Former colony $\times$ Year FE			✓	
Destination controls				✓
Observations	895,117	894,891	895,117	895,042

TABLE A.II: Effect across destinations - estimator robust to heterogeneous treatment effects

This table presents the results of estimating the average treatment effect on volume of exports to high-risk destinations. We use annual data from 2011 to 2018. We include time fixed effects as well as firm-product-destination fixed effects. We present two estimators. The first one is the two-way fixed effects estimator in which we use OLS with the fixed effect structure we described to compute the average treatment effect. For this estimator, we also report the sum of the positive weights assigned to each of the individual treatment effects. As shown in [De Chaisemartin and d'Haultfoeuille \(2020\)](#), the OLS estimator is a weighted average of the individual treatment effects but the weights are not all positive, even though they must add up to one. The second estimator is the estimator developed by [De Chaisemartin and d'Haultfoeuille \(2020\)](#), which is robust to the presence of heterogeneous treatment effects in our sample. For this second estimator, we also report the number of switchers, i.e. the number of observations which observe a change from untreated to treated (the treatment group). Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	TWFE	Robust estimator
High-risk $\times$ Post 2014	-0.266*** (0.023)	-0.087*** (0.021)
Year FE	✓	✓
Firm $\times$ Product $\times$ Destination FE	✓	✓
Observations	1,268,307	128,946
Number of switchers		79,819
Sum of positive weights	1.015	

TABLE A.III: Effect across destinations - estimator robust to multiple treatments

This table presents the results of estimating the average treatment effect on volume of exports to high-risk destinations. We use annual data from 2013 to 2018. We include time fixed effects as well as firm-product-destination fixed effects. We estimate the average treatment effect for each of the risk weight groups, using countries with the lowest risk weight (0.2) as the control group. We present two estimators. The first one is the two-way fixed effects estimator in which we use OLS with the fixed effect structure we described to compute the average treatment effect. The second estimator is the one proposed by [De Chaisemartin and d'Haultfoeuille \(2022\)](#), which is robust to the presence of multiple treatments and to heterogeneous treatment effects. This estimator uses the method developed in [De Chaisemartin and d'Haultfoeuille \(2022\)](#). Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	OLS	Robust estimator
Medium risk $\times$ Post 2014	0.089* (0.045)	0.276*** (0.067)
High risk $\times$ Post 2014	-0.054* (0.031)	0.087*** (0.030)
Very high risk $\times$ Post 2014	-0.310*** (0.024)	-0.276*** (0.067)
Year FE	✓	✓
Firm $\times$ Product $\times$ Destination FE	✓	✓
Observations	1,268,307	79,819

TABLE A.IV: Effects on entry and exit

This table presents the results of estimating equations (3) and (4), where the dependent variable is the entry rate and the exit rate, respectively. The entry rate is the ratio of the number of entrants at time t to the number of firms at time t and the exit rate is the ratio of number of firms exiting between $t - 1$ and t to the number of firms in time $t - 1$. We use annual data from 2011 to 2018. We present estimates for the average treatment effect decomposed across three groups of high-risk destinations: (1) those with a risk weight of 0.5, (2) those with a risk weight of 1, and (3) those with a risk weight of 1.5. We include product-year and destination fixed effects. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also present the average entry and exit rates for 2013. Errors are clustered by destination. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	Entry			Exit		
	(1)	(2)	(3)	(4)	(5)	(6)
Risk weight = 0.5 $\times$ Post 2014	-0.056** (0.018)	-0.060* (0.017)	-0.025** (0.015)	-0.005 (0.012)	-0.008 (0.013)	-0.026 (0.019)
Risk weight = 1 $\times$ Post 2014	-0.010 (0.010)	-0.008 (0.009)	-0.011 (0.011)	0.026** (0.011)	0.021* (0.011)	0.013 (0.011)
Risk weight = 1.5 $\times$ Post 2014	-0.025* (0.015)	-0.022 (0.015)	-0.023* (0.013)	0.053*** (0.014)	0.048*** (0.014)	0.020* (0.011)
Mean in 2013	.484	.484	.484	.273	.273	.273
Destination FE	✓	✓	✓	✓	✓	✓
Year FE	✓			✓		
Product FE	✓			✓		
Product $\times$ Year FE		✓	✓		✓	✓
Destination controls				✓		✓
Observations	202,950	202,494	198,192	176,184	175,617	172,103

TABLE A.V: Effect on entry - estimator robust to heterogeneous treatment effects

This table presents the results of estimating the average treatment effect on entry into high-risk destinations. We use annual data from 2011 to 2018. We include time fixed effects as well as product-destination fixed effects. We present two estimators. The first one is the two-way fixed effects estimator in which we use OLS with the fixed effect structure we described to compute the average treatment effect. For this estimator, we also report the sum of the positive weights assigned to each of the individual treatment effects. As shown in [De Chaisemartin and d'Haultfoeuille \(2020\)](#), the OLS estimator is a weighted average of the individual treatment effects but the weights are not all positive, even though they must add up to one. The second estimator is the estimator developed by [De Chaisemartin and d'Haultfoeuille \(2020\)](#), which is robust to the presence of heterogeneous treatment effects in our sample. For this second estimator, we also report the number of switchers, i.e. the number of observations which observe a change from untreated to treated (the treatment group). Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	TWFE	Robust estimator
High-risk $\times$ Post 2014	-0.041*** (0.010)	-0.041*** (0.011)
Year FE	✓	✓
Firm $\times$ Product $\times$ Destination FE	✓	✓
Observations	188,257	21,737
Number of switchers		11,604
Sum of positive weights	1.002	

TABLE A.VI: Effect on exit - estimator robust to heterogeneous treatment effects

This table presents the results of estimating the average treatment effect on exit into high-risk destinations. We use annual data from 2011 to 2018. We include time fixed effects as well as product-destination fixed effects. We present two estimators. The first one is the two-way fixed effects estimator in which we use OLS with the fixed effect structure we described to compute the average treatment effect. For this estimator, we also report the sum of the positive weights assigned to each of the individual treatment effects. As shown in [De Chaisemartin and d'Haultfoeuille \(2020\)](#), the OLS estimator is a weighted average of the individual treatment effects but the weights are not all positive, even though they must add up to one. The second estimator is the estimator developed by [De Chaisemartin and d'Haultfoeuille \(2020\)](#), which is robust to the presence of heterogeneous treatment effects in our sample. For this second estimator, we also report the number of switchers, i.e. the number of observations which observe a change from untreated to treated (the treatment group). Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

toprule	TWFE	Robust estimator
High-risk $\times$ Post 2014	0.042*** (0.011)	0.027*** (0.007)
Year FE	✓	✓
Firm $\times$ Product $\times$ Destination FE	✓	✓
Observations	168,616	19,805
Number of switchers		10,299
Sum of positive weights	1.002	

TABLE A.VII: Effect on entry - estimator robust to multiple treatments

This table presents the results of estimating the average treatment effect on entry into high-risk destinations. We use annual data from 2013 to 2018. We include time fixed effects as well as firm-product-destination fixed effects. We estimate the average treatment effect for each of the risk weight groups, using countries with low sovereign risk as the control group. We present two estimators. The first one is the two-way fixed effects estimator in which we use OLS with the fixed effect structure we described to compute the average treatment effect. The second estimator is the one proposed by [De Chaisemartin and d'Haultfoeuille \(2022\)](#), which is robust to the presence of multiple treatments and to heterogeneous treatment effects. This estimator uses the method developed in [De Chaisemartin and d'Haultfoeuille \(2022\)](#). Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	OLS	Robust estimator
Medium risk $\times$ Post 2014	-0.076*** (0.019)	0.036*** (0.017)
High risk $\times$ Post 2014	-0.018* (0.010)	0.033** (0.023)
Very high risk $\times$ Post 2014	-0.048*** (0.013)	-0.036*** (0.017)
Year FE	✓	✓
Firm $\times$ Product $\times$ Destination FE	✓	✓
Observations	188,257	11,604

TABLE A.VIII: Effect on exit - estimator robust to multiple treatments

This table presents the results of estimating the average treatment effect on exit into high-risk destinations. We use annual data from 2013 to 2018. We include time fixed effects as well as firm-product-destination fixed effects. We estimate the average treatment effect for each of the risk weight groups, using countries with low sovereign risk as the control group. We present two estimators. The first one is the two-way fixed effects estimator in which we use OLS with the fixed effect structure we described to compute the average treatment effect. The second estimator is the one proposed by [De Chaisemartin and d'Haultfoeuille \(2022\)](#), which is robust to the presence of multiple treatments and to heterogeneous treatment effects. This estimator uses the method developed in [De Chaisemartin and d'Haultfoeuille \(2022\)](#). Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	OLS	Robust estimator
Medium risk $\times$ Post 2014	-0.010 (0.010)	-0.005 (0.019)
High risk $\times$ Post 2014	0.028** (0.012)	0.006 (0.018)
Very high risk $\times$ Post 2014	0.055*** (0.014)	0.005 (0.019)
Year FE	✓	✓
Firm $\times$ Product $\times$ Destination FE	✓	✓
Observations	168,614	10,299

TABLE A.IX: Effects on trade credit

We use firm-year data in the 2011–2018 period. We consider four outcome variables: (1) the logarithm of accounts payable, (2) the logarithm of accounts receivables, (3) the logarithm of inventories, and (4) the logarithm of the cash conversion cycle. We split firms into two groups according to the share of exports in 2014 that are directed to high-risk countries, excluding firms for which the share is either zero or one. We define firms with a share above the median as the treated group and use firms with a share below the median as the control group. We present estimates for the average treatment effect. We include firm and year fixed effects. The firm controls include: log of capital (where capital is defined as PP&E), leverage (total liabilities as a share of total assets), the EBITDA-to-assets ratio, sales growth, the sales-to-assets ratio, labor productivity (defined as total sales over the number of workers), the ratio of current liabilities to total liabilities, the age of the firm squared, exports as a share of total sales, and exports as a share of total sales squared. Errors are clustered by firm. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	Payables	Receivables	Inventories	Cash conversion cycle
High exposure $\times$ Post 2014	-0.018 (0.022)	-0.036* (0.020)	-0.050* (0.027)	-0.040 (0.030)
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Firm controls	✓	✓	✓	✓
Number of firms	3,416	3,414	3,076	2,644
Observations	23,929	23,796	21,398	16,841

B Appendix to Section 5

TABLE B.I: Effects on number of loans

We use firm-year data in the 2011–2018 period and use the logarithm of the number of loans as the outcome variable. We split firms into two groups according to the share of exports in 2014 that are directed to high-risk countries, excluding firms for which the share is either zero or one. We define firms with a share above the median as the treated group and use firms with a share below the median as the control group. We present estimates for the average treatment effect. We include firm and year fixed effects. The time-varying firm controls include: log of total sales, the sales-to-asset ratio, the leverage ratio, the EBITDA-to-assets ratio, the growth rate of total sales, labor productivity, the ratio of current-to-total liabilities, the ratio of current-to-total assets, as well as the firm's age and its age squared. Errors are clustered by firm. *,** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	(1)	(2)	(3)	(4)
High exposure $\times$ Post 2014	-0.014 (0.026)	-0.012 (0.023)	-0.028 (0.026)	-0.027 (0.024)
Firm FE		✓		✓
Year FE		✓		✓
Controls			✓	✓
Number of firms	10,008	8,394	9,969	8,340
Observations	38,090	36,476	37,843	36,214

FIGURE B.1: Dynamic average treatment effect on loan conditions

This figure presents the results of estimating regression (5), where the dependent variable is the interest rate or the loan amount for loan k obtained by firm i from bank b in year t , and where we allow the average treatment effect over time. We use individual loan data for exporting firms from 2013 to 2018, and consider only loans with maturities under 180 days. We present estimates for the average treatment effects. The time-varying loan controls include: the loan maturity, the log of the loan amount and a dummy variable that takes the value of 1 if the loan is collateralized and 0 if otherwise. The time-varying firm controls include: log of total sales, the sales-to-asset ratio, the leverage ratio, the EBITDA-to-assets ratio, the growth rate of total sales, labor productivity, the ratio of current-to-total liabilities, the ratio of current-to-total assets, as well as the firm's age and its age squared. We include firm and bank-year fixed effects. Errors are clustered by firm and we present 95 percent confidence intervals.

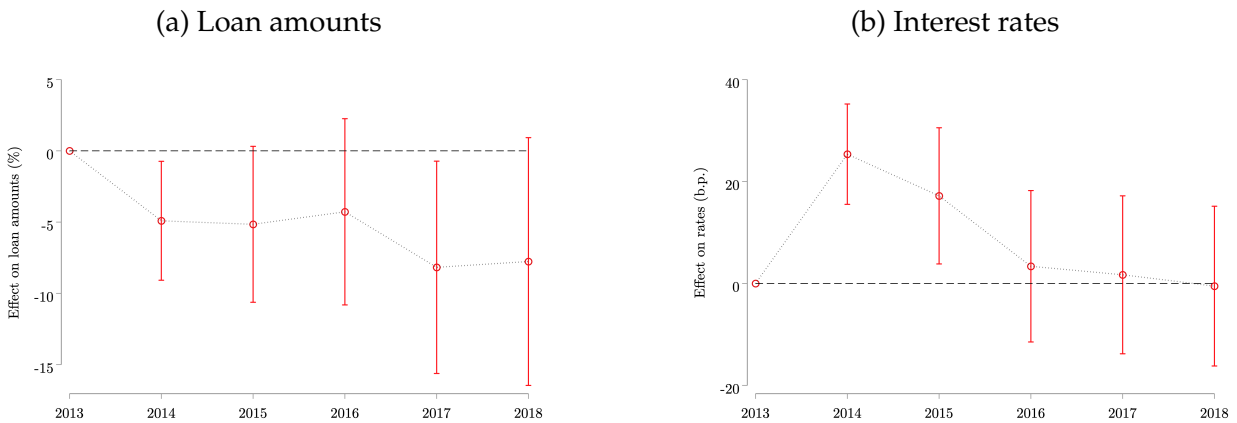
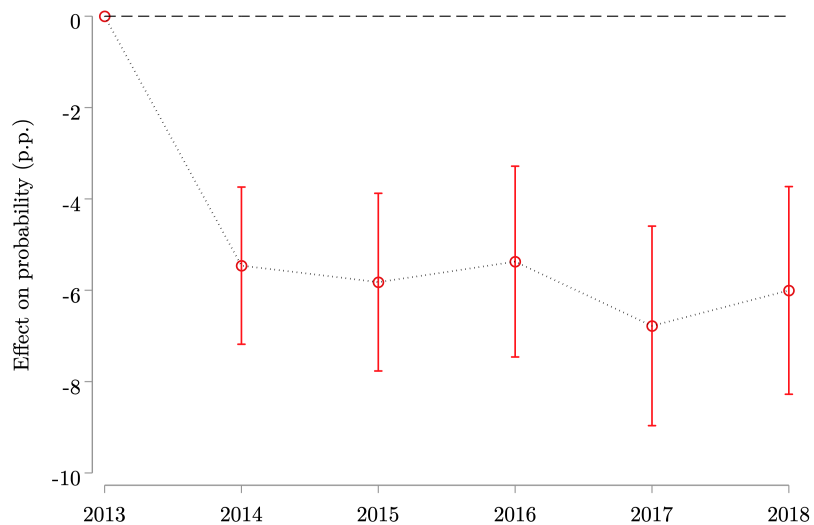


FIGURE B.2: Dynamic average treatment effects on probability of obtaining a loan

This figure presents the results of estimating regression (6), where the dependent variable is an indicator variable which takes the value of 1 if firm i receives at least one bank loan in year t and zero if otherwise. We use individual loan data for exporters from 2013 to 2018, and consider only loans with maturities under 180 days. We present estimates for the average treatment effect over time. The time-varying firm controls include: log of total sales, the sales-to-asset ratio, the leverage ratio, the EBITDA-to-assets ratio, the growth rate of total sales, labor productivity, the ratio of current-to-total liabilities, the ratio of current-to-total assets, as well as the firm's age and its age squared. Errors are clustered by firm and we present 95 percent confidence intervals.



C Appendix to Section 6

C.1 Data

This appendix describes the data sources we use in solving the model. We consider 31 countries: Argentina, Australia, Austria, Brazil, Canada, Chile, China, Denmark, Finland, France, Germany, Greece, Hungary, India, Indonesia, Ireland, Italy, Japan, Korea, Mexico, Netherlands, New Zealand, Norway, Portugal, South Africa, Spain, Sweden, Turkey, U.K., U.S., and a constructed rest of the world. We consider 40 sectors, which we report in Table C.I.

TABLE C.I: Tradable and non-tradable sectors

Tradable			Nontradable		
Number	Description	ISIC Rev. 4	Number	Description	ISIC Rev. 4
1	Agriculture	1–3	21	Electricity	35–39
2	Mining	5–9	22	Construction	41–43
3	Food	10–12	23	Retail	45–47
4	Textile	13–15	24	Hotels	55–56
5	Wood	16	25	Land transport	49
6	Paper	17–18	26	Water transport	50
7	Petroleum	19	27	Air transport	51
8	Chemicals	20–21	28	Auxiliary transport	52, 79
9	Plastic	22	29	Post and telecom	53, 61
10	Minerals	23	30	Finance	64–66
11	Basic Metals	24	31	Real estate	68
12	Metal products	25	32	Renting	77
13	Machinery n.e.c.	28	33	Computers	62–63, 95
14	Office	26	34	R&D	72
15	Electrical	27	35	Other business	69–71, 73–74, 80–82
16	Communication	58–60	36	Public admin	84
17	Medical	325, 266	37	Education	85
18	Auto	29	38	Health	75, 86–88
19	Other transport	30	39	Other services	78, 90–96
20	Other	31–33	40	Private households	97–98

In Figure C.1 we plot the share of exports which we can attribute to the thirty countries we include in our model. In 2013, which is our year of interest, these countries account for two thirds of world exports. This representativeness is larger for E.U. countries than it is for high-risk countries. However, even for high-risk countries we are able

TABLE C.II: Sectors according to dependence on credit

Number	Sector	Credit Group
1	Agriculture	Low credit
2	Mining	Low credit
3	Food	Low credit
4	Textile	High credit
5	Wood	Low credit
6	Paper	Low credit
7	Petroleum	Low credit
8	Chemicals	High credit
9	Plastic	Low credit
10	Minerals	Low credit
11	Basic Metals	High credit
12	Metal products	High credit
13	Machinery n.e.c.	High credit
14	Office	Low credit
15	Electrical	High credit
16	Communication	Low credit
17	Medical	High credit
18	Auto	High credit
19	Other transport	High credit
20	Other	High credit

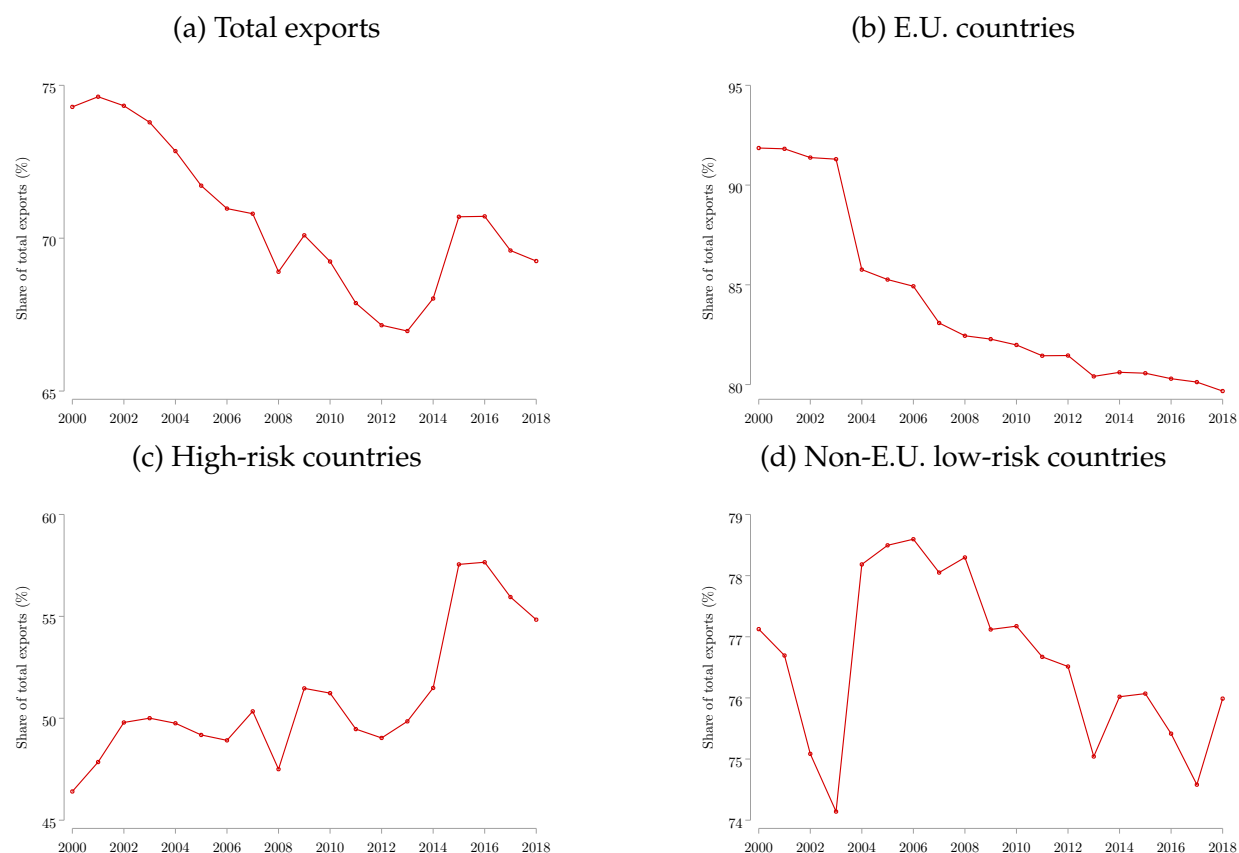
to include half of world exports. The share is smaller for high-risk countries because there are many more of these countries and, for most of them, there is not enough data on input- output tables to include them in the quantitative model. In Figure C.2, we do the same exercise but with imports. The results are very similar. In Figure C.3, we plot the share of exports and imports we attribute to the “Rest of the World” country which come from high-risk countries. As more than half of exports and imports in this fictional country come from high-risk countries, we will assign this fictional country to the group of high-risk countries.

C.1.1 Bilateral trade shares

We use bilateral trade flows for 2013 for our sample of countries and the first 20 sectors (which are the tradable sectors) in Table D.I. Bilateral trade data come from CEPII, which collects bilateral trade from the United Nations Statistical Division Commodity Trade database. Value are in thousands of dollars at current prices and exclude cost, insurance

FIGURE C.1: Model coverage - exports

This Figure presents the share of world exports attributable to the countries in our quantitative model. In panel (a), we present the share of world exports that is attributable to the countries in our quantitative model. In panel (b) we present the same share for E.U. countries. In panel (c), we present the same share for high-risk countries. In panel (d), we present the same share for non-E.U. low-risk countries.



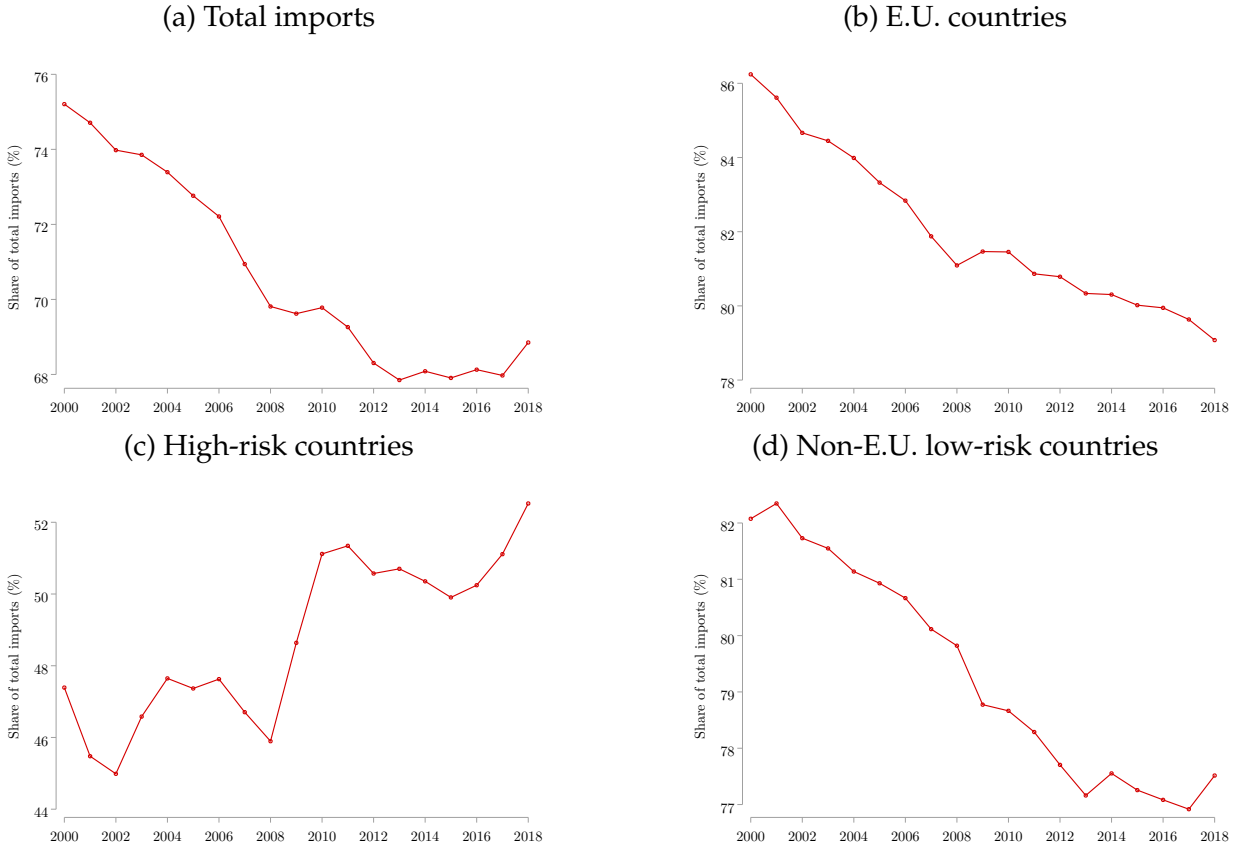
and freight (CIF). We define commodities using the Harmonized Commodity Description and Coding System (HS) 12. We match each commodity to a 2-digit ISIC Rev. 4 industry code using the OECD's concordance table. To compute imports from the rest of the world we, for each country in our sample, subtract total imports from all other countries in the sample from total imports of that country. To compute exports to the rest of the world we, for each country in our sample, subtract export to all other countries in the sample from total exports of that country.

C.1.2 Tariffs

We obtain bilateral trade tariffs at the sectoral level for the year 2013 from World Integrated Trade Solutions (WITS). We extract the effectively applied tariffs for all reporting

FIGURE C.2: Model coverage - imports

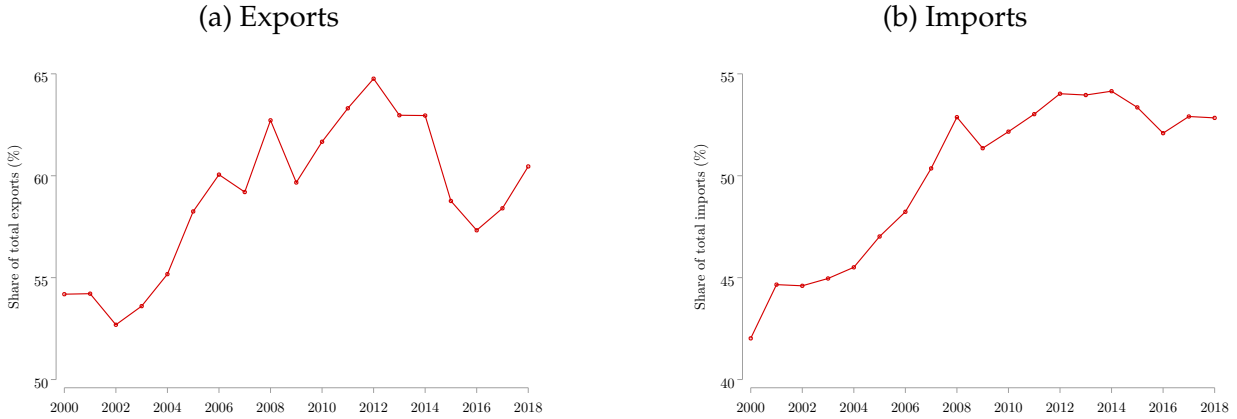
This Figure presents the share of world imports attributable to the countries in our quantitative model. In panel (a), we present the share of world imports that is attributable to the countries in our quantitative model. In panel (b) we present the same share for E.U. countries. In panel (c), we present the same share for high-risk countries. In panel (d), we present the same share for non-E.U. low-risk countries.



countries and partner countries and for all 3 digit ISIC Rev. 3 sectors. We then map the 3 digit ISIC Rev. 3 sectors to ISIC Rev. 4 sectors and then to our sectoral classification. We also use data from years 2011, 2012, 2014 and 2015 to fill some missing values for tariff data. At the end of this process, close to 20 percent of the observations still contain missing data. In order to address this we will use two sequential algorithms. First, we replace the missing value with the median tariff applied by the same country to the same source country (across all sectors). Second, we replace the missing values that are not replace by the first manual input with the median tariff applied by the same country to the same sector products (across all source countries). This allows us to fill all of the missing values.

FIGURE C.3: Rest of the World

This Figure presents the relevance of high-risk countries to the “Rest of the World” country. In panel (a), we report the share of exports of this group of countries which are from high-risk countries. In panel (b), we report the share of imports of this group of countries which are from high-risk countries.



C.1.3 Value added and gross production

Following [Caliendo and Parro \(2015\)](#), we obtain data on gross output and value added at the sectoral level for the year 2013 from three different data sets. We list these data sets in order of preference. For example, if we have data for the same variable for the same sector-country pair from data set 1 and from data set 2, we will use the value from data set 1.

OECD STAN: our first data set is the OECD STAN database for industrial analysis. This data set contains information on gross output and value added for OECD countries at the sectoral level using the ISIC Rev. 4 classification at current prices and in national currency. We convert these values to U.S. dollars using the exchange rates available at the OECD STAN database. This database allows us to fill around 75 percent of the information on value added per sector.

INDSTAT: our second data set is the Industrial Statistics Database INDSTAT2. This data set contains information at current prices in U.S. dollars for 71 3-digit manufacturing sectors. We aggregate these sectors at the 2-digit level and then use the allocation in Table D.I to compute value added and gross output for each sector-country. Using these two data sets allows us to fill most of the data for all countries in our sample.

OECD I-O and UNSTATS: we now need to compute the value added and gross production for the remaining sectors and countries. First, we use information from the OECD's Input-Output Tables to compute the value added and gross production for any sector and OECD country with missing data. Second, we use this data set to compute value added for the following large sectors: Agricultural, Hunting, Forestry and Fishing (sector 1), Mining and Utilities (sectors 2-12 and 21), Manufacturing (sectors 13-15 and 17-20), Construction (sector 22), Wholesale, retail trade, restaurants and hotels (sectors 23-24), Transport, storage and communication (sectors 16 and 25-29) and other activities (sectors 30-40). We add up value added across all countries. For each of these big sectors, we compute total value added and the share of each sector in value added of the big sector. We then obtain data on these big sectors from the United Nations National Accounts Database, which contains value added data for 200 countries. We decompose the value added of these big sectors into our sectoral classification using the shares we computed in the OECD I-O database. We then use the median sectoral share of value added in the OECD I-O database to compute gross production.

Rest of the world: we need to compute sectoral value added and gross output for the rest of the world. We begin by computing the world's value added for the big sectors using the United Nations Database, after excluding the countries in our sample. We then apply the sectoral shares of value added we obtained from the I-O database to split the value added in our 40 sectors and use the shares of value added in the OECD I-O database to compute gross production.

Filters: we also impose some filters in all stages of this process. We exclude observations where gross production or value added are negative. We also interpret observations where value added is larger than gross production as an error and therefore exclude them accordingly.

C.1.4 Input-Output tables and intermediate consumption

For each sector, we need information on the share of intermediate inputs per sector of origin. We obtain these data from the WIOD database, which contains I-O tables for 43 countries. We use information from 2013 and combine information on use of domestic intermediate inputs and imported intermediate inputs. This database allows us to compute the shares for most of the sector-countries in our sample. We fill missing values by taking the median share for that sector across countries.

C.1.5 Dispersion of productivity

We use the estimates for the dispersion of productivity from [Caliendo and Parro \(2015\)](#).

C.2 Identification

To study the effect of Basel III in exports from the E.U. to high-risk destinations, we use aggregate trade data from CEPII.³⁴ These data are at the exporter-importer-product-year level, where products are identified according to the 6-digit HS code. We aggregate the data to a 4-digit product classification and use focus on the 2008–2018 period. Our data set contains information on 18,175 destination-product pairs. We then estimate the following equation using data between 2008 and 2013:

$$\log \text{Exports}_{spt} = \mu_d + \lambda_{spt} + \beta X_{dt} + \gamma Z_{dt} + u_{spt}, \quad (\text{C.1})$$

We present the results of estimating equation (C.1) in Table C.III. We also conduct a triple difference exercise which we use to calibrate the model and which we present in Table C.IV.

³⁴The data is available at [CEPII](#). CEPII uses data from COMTRADE, after correcting for differences between import and export values.

TABLE C.III: Effect on exports to high-risk destinations for all E.U. countries

This table presents the results of estimating regression (C.1), where the dependent variable is the log of exports of country s of product p to destination d at time t . We use annual data from 2010 to 2018. We present estimates for the average treatment effect. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also include destination and source-product-year fixed effects. Errors are clustered by destination. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
High-risk $\times$ Post 2014	0.006** (0.003)	0.011*** (0.003)	-0.040*** (0.002)	-0.002 (0.003)	-0.048*** (0.003)
Year FE	✓				
Destination FE	✓	✓	✓	✓	✓
Source FE	✓				
Product FE	✓	✓		✓	
Year $\times$ Source FE		✓		✓	
Year $\times$ Source $\times$ Product FE			✓		✓
Destination controls				✓	✓
Observations	17,156,788	17,156,788	17,146,341	16,592,948	16,582,259

TABLE C.IV: Effect on exports to high-risk destinations for all E.U. countries -
decomposition by product dependence on credit

This table presents the results of estimating regression (C.1), where the dependent variable is the log of exports of country s of product p to destination d at time t . We use annual data from 2010 to 2018. We present estimates for the average treatment effect for products with low credit dependence and for the difference in average treatment effects between high and low credit dependence products. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also include destination and source-product-year fixed effects. We also present the share of exports of high-credit products within exports to high-risk countries. Errors are clustered by destination. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	(1)	(2)	(3)	(4)
High-risk $\times$ Post 2014	0.025 (0.031)	-0.023 (0.030)	0.016 (0.030)	-0.031 (0.029)
High-risk $\times$ High-credit $\times$ Post 2014	-0.036*** (0.011)	-0.032*** (0.010)	-0.037*** (0.011)	-0.032*** (0.010)
Share of high-credit products (%)	42	42	42	42
Destination FE	✓	✓	✓	✓
Year FE	✓		✓	
Source FE	✓		✓	
Product FE	✓		✓	
Source $\times$ Product $\times$ Year FE		✓		✓
Destination controls			✓	✓
Observations	17,156,788	17,146,341	16,592,948	16,582,259

Supplemental Appendix

Not for Publication

D Additional Empirical Results

FIGURE D.1: High-credit vs. low-credit products - product uses

This Figure presents the distribution of product uses for high- and low-credit products. We obtain product uses from the BEA classification of products into three groups: capital, intermediate goods and consumption goods. We assign products to the high-credit group or to the low-credit group according to their cash conversion cycle: products above the median are high-credit products and products below the median are low-credit products.

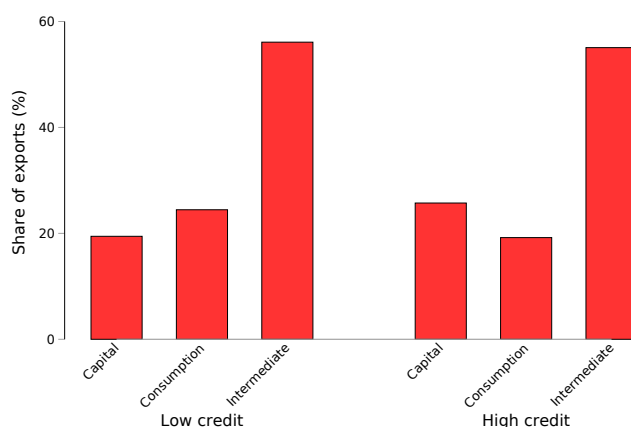


FIGURE D.2: High credit vs. low credit products - distribution of trade elasticities

This Figure presents the kernel distributions of the trade elasticities of high-credit and low-credit products. We obtain trade elasticities from CEPII and aggregate them at the 4-digit product level by taking the averages of the elasticities at the 6-digit level. We assign products to the high-credit group or to the low-credit group according to their cash conversion cycle: products above the median are high-credit products and products below the median are low-credit products. We consider only trade elasticities below 20 in order to improve the readability of the plot.

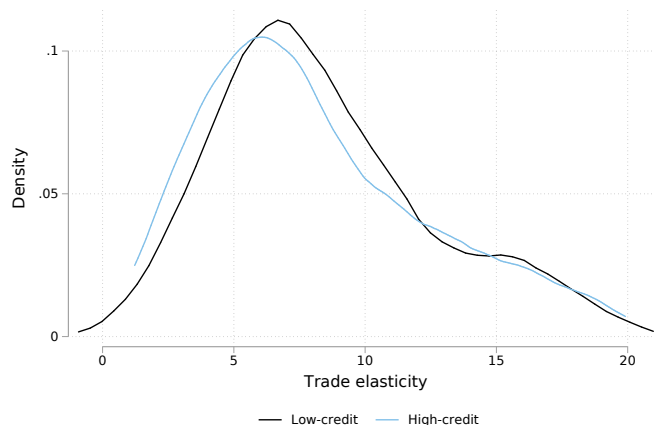


FIGURE D.3: Share of high-credit and low-credit products in exports

This Figure presents the decomposition of total exports in exports of high-credit and low-credit products. We assign products to the high-credit group or to the low-credit group according to their cash conversion cycle: products above the median are high-credit products and products below the median are low-credit products. In panel (a) we present this decomposition over time for total Portuguese exports and in panel (b) we consider exports to high-risk destinations.

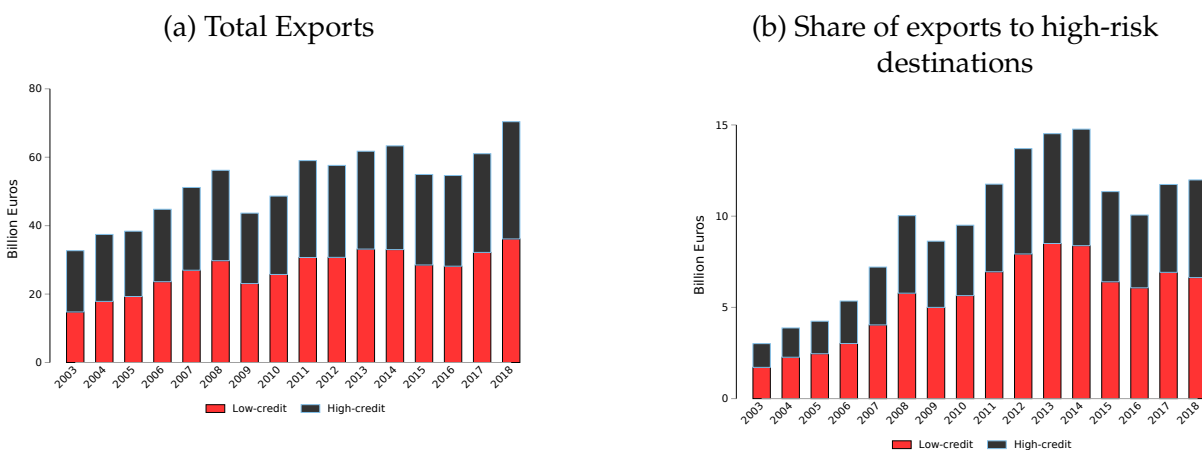


TABLE D.I: Effect on exports to high-risk destinations from non-E.U. countries

This table presents the results of estimating regression (C.1) for non-E.U. countries and where the dependent variable is the log of exports of country s of product p to destination d at time t . We use annual data from 2010 to 2018 for the exports of non-E.U. countries. We present estimates for the average treatment effect. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also include destination and source-product-year fixed effects. Errors are clustered by destination. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
High-risk $\times$ Post 2014	0.136** (0.063)	0.120** (0.060)	0.082 (0.050)	0.093 (0.064)	0.078 (0.061)	0.034 (0.050)
Share of high-risk (%)	48	48	48	48	48	48
Destination FE	✓	✓	✓	✓	✓	✓
Year FE	✓			✓		
Exporter FE	✓			✓		
Product FE	✓	✓		✓	✓	
Source $\times$ Year FE		✓			✓	
Source $\times$ Product $\times$ Year FE			✓			✓
Destination controls				✓	✓	✓
Observations	26,925,468	26,925,468	26,604,900	26,132,221	26,132,221	25,810,296

TABLE D.II: Effect on exports to high-risk destinations around the Great Recession

This table presents the results of estimating regression (C.1) for E.U. countries and where the dependent variable is the log of exports of country s of product p to destination d at time t . We use annual data from 2003 to 2013. We present estimates for the average treatment effect: we compare exports to high-risk vs. exports to low-risk countries, before and after 2008. The time-varying destination controls include the log of GDP, the logarithm of total exports, the share of imports to GDP, the nominal exchange rate, the real exchange rate, the share of the current account to GDP and the sovereign risk rating, as well as region-year fixed effects. We also include destination and source-product-year fixed effects. Errors are clustered by destination. *, ** and *** denote significance at the 10%, 5% and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
High-risk $\times$ Post 2008	0.143*** (0.042)	0.155*** (0.041)	0.142*** (0.043)	-0.027 (0.037)	-0.020 (0.036)	-0.042 (0.037)
Share of high-risk (%)	24	24	24	24	24	24
Destination FE	✓	✓	✓	✓	✓	✓
Year FE	✓			✓		
Exporter FE	✓			✓		
Product FE	✓	✓		✓	✓	
Source $\times$ Year FE		✓			✓	
Source $\times$ Product $\times$ Year FE			✓			✓
Destination controls				✓	✓	✓
Observations	15,052,396	15,052,396	15,041,476	14,546,122	14,546,122	14,534,976

E Additional Model Results

E.1 Model setup

We consider a multi-country multi-sector Ricardian model of international trade as in [Caliendo and Parro \(2015\)](#). The model is static. There are N countries indexed by either n or i and, within each country, there are J sectors indexed by j or k . Throughout the description of the model, subscripts will always denote countries and superscripts will always denote sectors. In this model, trade exists because firms want to purchase the cheapest inputs for production. Households will not consume foreign goods. There are two types of goods within each sector: composite goods and intermediate goods. Composite goods use intermediates from their sector in their production and are non-tradable. Producers of composite goods may import intermediates. Intermediate goods use domestic composite goods from all sectors and labor in production and may be tradable.

E.1.1 Households and preferences

In each country, there is a representative household with preferences

$$U(C_n) = C_n = \prod_{j=1}^J (C_n^j)^{\alpha_n^j}, \quad \sum_{j=1}^J \alpha_n^j = 1,$$

where C_n^j is the consumption of sector j composite goods by the representative household and C_n is aggregate consumption. The share parameters $\alpha_n^j \geq 0$ may vary by country and capture the share of expenditure by the representative household on each sector. The household only consumes domestic goods, i.e. C_n^j is produced domestically. Given this structure, the domestic CPI is given by

$$P_n = \prod_{j=1}^J \left(\frac{P_n^j}{\alpha_n^j} \right)^{\alpha_n^j},$$

where P_n^j is the price of the composite good j in country n .

The household has two sources of income. First, the representative household is endowed with L_n units of labor which it supplies inelastically at a wage w_n .³⁵ Second, the household receives a lump-sum transfer from the government with the proceeds from

³⁵We are therefore assuming that there is free labor mobility across sectors within a particular country but that there is no labor mobility across countries.

tariffs imposed on imports.

E.1.2 Production

Intermediate goods: in each sector-country (j, n) there is a continuum of intermediate goods $\omega^j \in [0, 1]$, where each variety ω^j exists in all countries. Production requires two types of factors: labor and materials. Materials are composite goods from all sectors in the country. Therefore, an intermediate producer in country n and sector j may use composite goods from all sectors in country n . Each intermediate producer has a productivity $z_n^j(\omega^j)$. Production takes the form

$$q_n^j(\omega^j) = z_n^j(\omega^j) \left(l_n^j(\omega^j) \right)^{\gamma_n^j} \prod_{k=1}^J \left(m_n^{kj}(\omega^j) \right)^{\gamma_n^{kj}}, \quad \gamma_n^j + \sum_{k=1}^J \gamma_n^{kj} = 1$$

where l_n^j is labor and m_n^{kj} are materials from sector k used in production. The parameter $\gamma_n^j \geq 0$ represents the share of value added in production and γ_n^{kj} represents the share of production that is allocated to inputs from sector k . The factors shares and the share of value added may vary across sectors and countries.

Markets are perfectly competitive and so producers of intermediate goods will price their goods at the marginal cost which is $c_n^j / z_n^j(\omega^j)$ where

$$c_n^j = Y_n^j w_n^{\gamma_n^j} \prod_{k=1}^J \left(p_n^k \right)^{\gamma_n^{kj}}, \quad (\text{E.1})$$

and where $Y_n^j = \left(\gamma_n^j \right)^{-\gamma_n^j} \prod_{k=1}^J \left(\gamma_n^{kj} \right)^{-\gamma_n^{kj}}$ is a constant that varies across sectors and countries. We will interpret c_n^j as the unit cost of the production bundle.

Composite goods: producers of composite goods in sector j and country n supply a quantity Q_n^j according to the production function

$$Q_n^j = \left[\int_0^1 \left(h_n^j(\omega^j) \right)^{1-1/\sigma^j} d\omega^j \right]^{\sigma^j / (\sigma^j - 1)}, \quad \sigma > 0$$

where $\sigma > 0$ is the elasticity of substitution across intermediate goods and $h_n^j(\omega^j)$ is the demand for intermediate good ω^j by composite good producer j in country n . Cost

minimization yields the following demand for intermediates

$$h_n^j(\omega^j) = \left(\frac{p_n^j(\omega^j)}{P_n^j} \right)^{-\sigma^j} Q_n^j,$$

where $p_n^j(\omega^j)$ is the price of the intermediate good and P_n^j is the unit price of the composite good which is given by

$$P_n^j = \left[\int_0^1 \left(p_n^j(\omega^j) \right)^{1-\sigma^j} d\omega^j \right]^{\frac{1}{1-\sigma^j}}.$$

Producers of composite goods may purchase intermediate ω^j from any country and they choose the lowest cost supplier.

E.1.3 Trade costs and prices

Prices of intermediate goods: producers of composite goods may purchase intermediate goods from any country, and they choose the lowest cost supplier. In the presence of trade costs, a unit of an intermediate good ω^j produced in country i is available in country n at a price $c_i^j \kappa_{in}^j / z_i^j(\omega^j)$. Therefore, the cost to the composite good producer of purchasing the lowest price intermediate ω^j is

$$p_n^j(\omega^j) = \min_i \left\{ \frac{c_i^j \kappa_{in}^j}{z_i^j(\omega^j)} \right\}.$$

If sector j is a non-tradable sector, we set $\kappa_{in}^j = \infty$ for all $i \neq n$ and this implies that $p_n^j(\omega^j) = c_n^j / z_n^j(\omega^j)$ as the lowest cost supplier is the domestic producer.

We adopt a probabilistic representation of productivities. In particular, we assume that the productivity of producing an intermediate good ω^j in country n is the realization of a Fréchet distribution with a location parameter $\lambda_n^j \geq 0$ and a shape parameter $\theta^j \geq 0$. We also assume that the distribution of productivities is independent across goods, sector and countries and that $1 + \theta^j \geq \sigma^j$. This representation allows to independently vary absolute and comparative advantages. The location parameter λ_n^j represents absolute advantage, as a higher λ_n^j increases average productivity. The shape parameter θ^j represents comparative advantage as a smaller θ^j implies a higher dispersion of productivities. With

this distributional assumption, we can then compute the sectoral price index as

$$P_n^j = A^j \left[\sum_{i=1}^N \lambda_i^j \left(c_i^j \kappa_{in}^j \right)^{-\theta^j} \right]^{-1/\theta^j}, \quad (\text{E.2})$$

where A^j is a constant. For a non-tradable sector where $\kappa_{in}^j = \infty$ for all $i \neq n$, $P_n^j = A^j \left(\lambda_n^j \right)^{-1/\theta^j} c_n^j$.

E.1.4 Expenditure shares and market clearing

We can define total expenditure on sector j goods in country n as X_n^j . Define X_{in}^j as the expenditure in country n of goods from sector j coming from country i . Using these two terms, we can write the share of country i in country n 's expenditure of sector j goods as $\pi_{in}^j = X_{in}^j / X_n^j$. Using the properties of the Fréchet distribution, we can then express this share as a function of unit costs, trade costs and exogenous parameters as

$$\pi_{in}^j = \frac{\lambda_i^j \left[c_i^j \kappa_{in}^j \right]^{-\theta^j}}{\sum_{h=1}^N \lambda_h^j \left[c_h^j \kappa_{hn}^j \right]^{-\theta^j}}, \quad (\text{E.3})$$

and for a non-tradable sector where $\kappa_{in}^j = \infty$ for all $i \neq n$, $\pi_{nn}^j = 1$. With this result, we can now write the gravity equation of this model:

$$X_{in}^j = \frac{\lambda_i^j \left[c_i^j \kappa_{in}^j \right]^{-\theta^j}}{\sum_{h=1}^N \lambda_h^j \left[c_h^j \kappa_{hn}^j \right]^{-\theta^j}} X_n^j, \quad (\text{E.4})$$

where trade flows depend on demand in the destination country X_n^j as well as productivity across all possible countries (λ_h^j), marginal costs across all countries (c_h^j), trade costs κ_{hn}^j and the dispersion of the distribution of productivities θ^j . This term appears in the same fashion as trade elasticities appear in Armington models. We will therefore often refer to the dispersion of the distribution of productivity as the trade elasticity.

We now turn to market clearing. Consider sector j in country n with a value of production X_n^j . This production is used both for final consumption of households in country n and as materials by intermediate producers in country n . Given the Cobb-Douglas assumption we made for the utility function of the representative household, production used for household consumption is $\alpha_n^j I_n$, where I_n is the income of the representative

household. From the Cobb-Douglas assumption in the production function, each intermediate good producer in sector k will use a share γ_n^{jk} of its production to purchase composite goods from sector j . In turn, total production of these intermediate goods will be sold to all sectors k in all countries. Therefore, we can write the market clearing condition as

$$X_n^j = \alpha_n^j I_n + \sum_{k=1}^J \gamma_n^{jk} \sum_{i=1}^N X_i^k \frac{\pi_{ni}^k}{1 + \tau_{ni}^k}, \quad (\text{E.5})$$

where $\sum_{i=1}^N X_i^k \pi_{ni}^k / (1 + \tau_{ni}^k)$ represents total production of intermediates in sector k in country n . Note that in equation (E.5), we are removing tariffs from total expenditure. We do this because firms only receive the unit costs, not the tariffs. We don't do this for iceberg costs or interest rates because the value of these frictions is a loss of resources and is not distributed to households.

Households derive income from two sources: (1) labor income $w_n L_n$, where w_n is the wage and L_n is the exogenous labor endowment in country n and (2) tariff revenue R_n . Therefore, household income is given by $I_n = w_n L_n + R_n$. As tariffs are imposed on all tariffs, tariff revenue can be written as

$$R_n = \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j X_n^j \frac{\pi_{in}^j}{1 + \tau_{in}^j},$$

which is just the sum of the value of the ad-valorem flat rate tariffs imposed on unit costs (which is why we need to divide the shares π_{in}^j by the tariffs). We assume national trade deficits are zero but sectoral trade deficits are endogenous. National trade deficits are given by $D_n = \sum_{j=1}^J D_n^j$, where D_n is exogenous but D_n^j is an equilibrium object. Sectoral deficits are given by

$$D_n^j = \sum_{i=1}^N X_n^j \frac{\pi_{in}^j}{1 + \tau_{in}^j} - \sum_{i=1}^N X_i^j \frac{\pi_{ni}^j}{1 + \tau_{ni}^j}, \quad (\text{E.6})$$

which is the difference between imports and exports. We can use the definition of national deficits together with (E.6) to write the balanced trade equation

$$\sum_{j=1}^J \sum_{i=1}^N X_n^j \frac{\pi_{in}^j}{1 + \tau_{in}^j} = \sum_{j=1}^J \sum_{i=1}^N X_i^j \frac{\pi_{ni}^j}{1 + \tau_{ni}^j}, \quad (\text{E.7})$$

in which the left hand side is the value of exports and the right hand side is the value of imports.³⁶

³⁶We have not specified market clearing for the labor market. We don't need to do this because market

E.1.5 Equilibrium

We can now define an equilibrium under policies $\{r_{ni}^j, \tau_{ni}^j\}$.

Definition 1 Given $L_n, D_n, \lambda_n^j, \theta^j$ and d_{ni}^j , and equilibrium under policy $\{r, \tau\}$ is a wage vector $w \in \mathbb{R}_{++}^N$ and prices $\{P_n^j\}_{j=1, n=1}^{J, N}$ that satisfy equilibrium conditions (E.1), (E.2), (E.3), (E.5) and (E.7) for all j, n .

Instead of solving for an equilibrium under a policy $\{r, \tau\}$ and then solving for another equilibrium under a new policy $\{r', \tau\}$, we will use the exact hat algebra method of Dekle et al. (2008) to solve for the equilibrium in relative changes.³⁷ In this method, instead of computing two equilibria and then computing the changes, we can compute the changes directly. This method is appealing for two reasons. First, we can identify the effect on equilibrium outcomes from a pure change in the cost of credit. Second, we can solve the model without needing to estimate parameters which may be difficult to identify. For example, this method implies that we do not need to specify the interest rates before Basel III and we can instead focus on the change in interest rates.

Therefore, if we have a change in interest rates we can then compute the equilibrium changes without relying on estimates of productivity or transport costs. We only need data on bilateral trade shares π_{in}^j , the share of value added in production γ_n^j , value added $w_n L_n$, the share of intermediate consumption γ_n^{kj} and the sectoral dispersion of productivity θ^j . We will obtain the share of each sector in final demand α_n^j from these data.

E.1.6 Aggregation

In this model, we can write a sectoral production function as a function of factor usage in the sector. Total production in sector j in country n can be written as

$$\frac{Y_n^j}{P_n^j} = \frac{c_n^j}{P_n^j} (L_n^j)^{\gamma_n^j} \prod_{k=1}^J (M_n^{kj})^{\gamma_n^{kj}},$$

clearing in each sector-country together with the balanced trade equation yields market clearing for the labor market in each country. In order to see this, we can add equation (E.5) across all sectors, use the expression for household income and then substitute into the balanced trade equation (E.6) to obtain

$$w_n L_n = \sum_{j=1}^J \gamma_n^j \sum_{i=1}^N X_i^j \frac{\pi_{ni}^j}{1 + \tau_{ni}^j}.$$

³⁷We present a formal definition of the equilibrium in changes in Appendix C.

where $Y_n^j = \int \left(c_n^j / z_n^j(\omega^j) \right) q_n^j(\omega^j) d\omega^j$ is the value of production in sector j and country n , L_n^j is total labor used in the sector and M_n^{kj} is total usage of materials from sector k in sector n . According to this representation, we can think of the ratio c_n^j / P_n^j as a multiplicative TFP factor. In fact, this term captures gains from trade in this model. To see this, note that in autarky, $d \log P_n^j = d \log c_n^j$ and so changes in marginal costs of domestic production move one-to-one with changes in price of the good j in country n . With trade, an increase in marginal costs no longer causes the same increase in the price, as this term will be dampened by the change in the share of own consumption π_{nn}^j . Therefore, in the presence of trade, increases in the cost of domestic production do not fully pass-through into the cost of the good in country n . Therefore, we shall call this ratio $A_n^j = c_n^j / P_n^j$ and will interpret it as productivity in sector j in country n . We can further use equation (E.3) to write changes in productivity as

$$d \log A_n^j = -\frac{1}{\theta^j} d \log \pi_{nn}^j, \quad (\text{E.8})$$

which implies that we only need to know two quantities to identify changes in productivity: (1) the trade elasticity and (2) the share of own consumption.³⁸ Suppose that there is a shock that increases the share of own consumption. In that case, marginal costs increase because firms are no longer importing cheap intermediates from abroad and are instead relying on the more expensive domestically produced intermediates. Therefore, production becomes more expensive and productivity decreases. This representation of gains from trade as a function of these two quantities is very general. In fact, [Arkolakis et al. \(2012\)](#) show that in most trade models gains from trade can be summarized by the elasticity of imports with respect to variable trade costs (which in this model is the trade elasticity) and the share of expenditure on domestic goods. In their seminal paper, the Ricardian model written by [Eaton and Kortum \(2002\)](#) displays this characterization, as does the [Caliendo and Parro \(2015\)](#) model.

E.2 Definitions

Definition 2 Let (w, P) be an equilibrium under policy $\{r, \tau\}$ and let (w', P') be an equilibrium under policy $\{r', \tau\}$. Define $(\hat{w}, \hat{P})$ as an equilibrium under policy $\{r', \tau\}$ relative to $\{r, \tau\}$, where for a variable x $\hat{x} = x' / x$.³⁹ Using equations (E.1), (E.2), (E.3), (E.5) and (E.7) the equilibrium

³⁸Note that, for non-tradable goods, $\pi_{nn}^j = 1$ and so productivity is fixed at $A_n^j = (A^j)^{-1} \left(\lambda_n^j \right)^{1/\theta^j}$.

³⁹Note that this definition can be easily extended to include changes in tariffs as well as changes in interest rates. We do not do so for the interest of simplicity.

conditions in relative changes satisfy:

Cost of the input bundles:

$$\hat{c}_n^j = (\hat{w}_n)^{\gamma_n^j} \prod_{k=1}^J \left(\hat{P}_n^k \right)^{\gamma_n^{kj}}. \quad (\text{E.9})$$

Price index:

$$\hat{P}_n^j = \left[\sum_{i=1}^N \pi_{in}^j \left(\hat{\kappa}_{in}^j \hat{c}_i^j \right)^{-\theta^j} \right]^{-1/\theta^j}. \quad (\text{E.10})$$

Bilateral trade shares:

$$\hat{\pi}_{in}^j = \left(\frac{\hat{c}_i^j \hat{\kappa}_{in}^j}{\hat{P}_n^j} \right)^{-\theta^j}. \quad (\text{E.11})$$

Total expenditure in each country n and sector j :

$$X_n^{j'} = \alpha_n^j I_n' + \sum_{k=1}^J \gamma_n^{jk} \sum_{i=1}^N X_i^{k'} \frac{\pi_{ni}^{k'}}{1 + \tau_{ni}^k}. \quad (\text{E.12})$$

Trade balance:

$$\sum_{j=1}^J \sum_{i=1}^N X_n^{j'} \frac{\pi_{in}^{j'}}{1 + \tau_{in}^j} - D_n = \sum_{j=1}^J \sum_{i=1}^N X_i^{j'} \frac{\pi_{ni}^{j'}}{1 + \tau_{ni}^j}, \quad (\text{E.13})$$

where $\hat{\kappa}_{in}^j = (1 + r_{in}^{j'}) / (1 + r_{in}^j)$ and $I_n' = \hat{w}_n w_n L_n + D_n + \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j X_n^{j'} \frac{\pi_{in}^{j'}}{1 + \tau_{in}^j}$.

E.3 Proofs

E.3.1 Proof of Proposition 1

The change in welfare is given by

$$d \log W_n = \frac{w_n L_n}{I_n} d \log w_n + \frac{R_n}{I_n} d \log R_n - d \log P_n.$$

From equation (E.9), the change in marginal costs for a given sector j in country n is given by

$$d \log c_n^j = \gamma_n^j d \log w_n + \sum_{k=1}^J \gamma_n^{kj} d \log P_n^k$$

and so the change in the wage is given by

$$d \log w_n = \frac{1}{\gamma_n^j} d \log c_n^j - \sum_{k=1}^J \frac{\gamma_n^{kj}}{\gamma_n^j} d \log P_n^k.$$

Using the market clearing condition for labor, it also follows that

$$\begin{aligned}
\frac{w_n L_n}{I_n} d \log w_n &= \frac{1}{I_n} \sum_{j=1}^N \gamma_n^j \sum_{i=1}^N E_{ni}^j d \log w_n \\
&= \frac{1}{I_n} \sum_{j=1}^N \gamma_n^j \sum_{i=1}^N E_{ni}^j \left\{ \frac{1}{\gamma_n^j} d \log c_n^j - \sum_{k=1}^J \frac{\gamma_n^{kj}}{\gamma_n^j} d \log P_n^k \right\} \\
&= \frac{1}{I_n} \sum_{j=1}^N \sum_{i=1}^N E_{ni}^j d \log c_n^j - \frac{1}{I_n} \sum_{j=1}^N \sum_{i=1}^N E_{ni}^j \sum_{k=1}^J \gamma_n^{kj} d \log P_n^k \\
&= \underbrace{\frac{1}{I_n} \sum_{j=1}^N \sum_{i=1}^N E_{ni}^j d \log c_n^j}_{A_1} - \underbrace{\frac{1}{I_n} \sum_{j=1}^N \sum_{k=1}^J \gamma_n^{kj} d \log P_n^k \sum_{i=1}^N E_{ni}^j}_{A_2}
\end{aligned}$$

where $E_{ni}^j \equiv X_i^j \pi_{ni}^j / (1 + \tau_{ni}^j)$ are the exports of sector j goods from country n to country i .

The change in tariff revenue is given by

$$\begin{aligned}
d \log R_n &= \frac{\sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j (d \log X_n^j + d \log \pi_{in}^j)}{R_n} \\
&= \frac{\sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j d \log M_{in}^j}{R_n}
\end{aligned}$$

where $M_{in}^j \equiv X_n^j \pi_{in}^j / (1 + \tau_{in}^j)$ are country n 's imports of sector j goods from country i . Therefore, it follows that

$$\frac{R_n}{I_n} d \log R_n = \underbrace{\frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j d \log M_{in}^j}_{A_3}.$$

From equation (E.10), we can write the change in the domestic CPI as

$$d \log P_n = \sum_{j=1}^J \alpha_n^j \sum_{i=1}^N \pi_{in}^j (d \log c_i^j + d \log \kappa_{in}^j)$$

and then, using the market clearing condition for sector j , we can write

$$\alpha_n^j = \frac{X_n^j}{I_n} - \frac{1}{I_n} \sum_{k=1}^J \gamma_n^{jk} \sum_{i=1}^N E_{ni}^k$$

and so the change in the domestic CPI is given by

$$\begin{aligned}
d \log P_n &= \sum_{j=1}^J \left\{ \frac{X_n^j}{I_n} - \frac{1}{I_n} \sum_{k=1}^J \gamma_n^{jk} \sum_{i=1}^N E_{ni}^k \right\} \sum_{i=1}^N \pi_{in}^j \left(d \log c_i^j + d \log \kappa_{in}^j \right) \\
&= \underbrace{\sum_{j=1}^J \frac{X_n^j}{I_n} \sum_{i=1}^N \pi_{in}^j \left(d \log c_i^j + d \log \kappa_{in}^j \right)}_{B_1} \\
&\quad - \underbrace{\frac{1}{I_n} \sum_{j=1}^J \sum_{k=1}^J \gamma_n^{jk} \sum_{i=1}^N E_{ni}^k \sum_{i=1}^N \pi_{in}^j \left(d \log c_i^j + d \log \kappa_{in}^j \right)}_{A_4}
\end{aligned}$$

and we can write the first term as

$$\begin{aligned}
B_1 &= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \pi_{in}^j X_n^j \left(d \log c_i^j + d \log \kappa_{in}^j \right) \\
&= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(1 + \tau_{in}^j \right) M_{in}^j \left(d \log c_i^j + d \log \kappa_{in}^j \right) \\
&= A_5 + A_6 + A_7
\end{aligned}$$

where

$$\begin{aligned}
A_5 &= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N M_{in}^j d \log c_i^j, \\
A_6 &= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(1 + \tau_{in}^j \right) M_{in}^j d \log \kappa_{in}^j, \\
A_7 &= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j d \log c_i^j.
\end{aligned}$$

Therefore, the change in the real wage can be written as

$$d \log W_n = A_1 - A_2 + A_3 + A_4 - A_5 - A_6 - A_7.$$

Gathering terms, note that

$$A_1 - A_5 = \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(E_{ni}^j d \log c_n^j - M_{in}^j d \log c_i^j \right)$$

$$A_3 - A_7 = \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j \left(d \log M_{in}^j - d \log c_i^j \right)$$

$$A_4 - A_2 = 0$$

$$-A_6 = -\frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(1 + \tau_{in}^j \right) M_{in}^j d \log \kappa_{in}^j$$

and so the change in welfare is given by

$$\begin{aligned} d \log W_n &= \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(E_{ni}^j d \log c_n^j - M_{in}^j d \log c_i^j \right) + \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \tau_{in}^j M_{in}^j \left(d \log M_{in}^j - d \log c_i^j \right) \\ &\quad - \frac{1}{I_n} \sum_{j=1}^J \sum_{i=1}^N \left(1 + \tau_{in}^j \right) M_{in}^j d \log \kappa_{in}^j \end{aligned}$$

as we wanted to show.

E.4 Additional results

E.4.1 Effect on exports

We start by looking at the direct effects of the shock on exports. In Table E.I we present the growth rate of the exports of E.U. exports, as well as the growth rates of exports to high-risk destinations and to low-risk destinations.

As expected, exports to high-risk countries decrease and the magnitudes are similar to our empirical analysis in Section ?? . For example, exports from Germany to high-risk countries decrease by 4.2 percent in response to an increase in interest rates faced by firms exporting to high-risk countries. Furthermore, as we discussed before, the model predicts a modest increase in exports to low-risk countries. This increase is driven by the fact that exports from high-risk countries have now become more expensive. This increase in cost is driven by an increase in the price of imports which then leads to an increase in marginal costs. However, this increase is not enough to prevent overall exports from E.U. countries to decrease. It's also important to note that is significant heterogeneity across E.U. countries in terms of impact on exports. For example, Dutch exports to high-risk countries fall by only 2.04 percent while Greek exports to high-risk countries fall by 5.54 percent. There are two factors that explain this heterogeneity. First, as the increase in interest rates is larger for high-credit sectors, countries with an export mix skewed towards these sectors will face a larger effective increase in interest rates. The second factor is related to trade elasticities. From the gravity equation in (18) we see that the

TABLE E.I: Effect on E.U. exports

This table presents the effect of the increase in interest rates on exports of E.U. countries. We present the growth rate of total exports, as well as the growth rates of exports to high-risk destinations and to low-risk destinations. We focus on the E.U. countries we include in our model: Austria, Denmark, Germany, Finland, France, Greece, Hungary, Ireland, Italy, Netherlands, Portugal, Spain, Sweden and the U.K.

	To high-risk countries	To low-risk countries	Total
Austria	-4.04	0.83	-0.97
Denmark	-3.89	0.48	-0.69
Finland	-4.12	0.86	-1.25
France	-4.45	1.24	-1.18
Germany	-4.31	1.06	-1.29
Greece	-5.74	1.92	-2.67
Hungary	-4.00	0.89	-1.14
Ireland	-3.14	0.71	-0.46
Italy	-4.31	0.76	-1.44
Netherlands	-2.13	0.51	-0.88
Portugal	-4.55	0.95	-0.88
Spain	-4.15	0.72	-1.25
Sweden	-4.24	1.04	-0.88
U.K.	-4.38	1.54	-1.19

impact on exports will increase (in absolute value) with the value of trade elasticities, as more elastic products respond more to the same increase in interest rates. Therefore, E.U. countries that focus on sectors with higher trade elasticities will face a larger drop in exports to high-risk destinations. We can analyze the impact of these two forces on E.U. exports by plotting a binned scatter plot in Figure 14 of exports by sector and country against trade elasticities for high-credit and low-credit sectors.

For a given level for the trade elasticity, the impact on exports of high-credit products is larger than the impact on exports of low-credit sectors. This follows from the fact that the shock on interest rates for high-credit products is larger than the shock on exports of low-credit sectors. As the trade elasticity increases, so does the impact on exports, which means that countries that mostly exported less elastic products faced a smaller decrease in exports. For example, sectors like Petroleum, which has a trade elasticity of 65, observed very large drops in exports (close to 25 percent), whereas sectors like Machinery which has a trade elasticity of 1 face smaller drops in exports.

This heterogeneity suggests that the impact of this increase in interest rates has very

different impacts on different sectors and countries. For example, for a low-credit sector with a low trade elasticity, the impact is very close to zero. For a high-credit sector with a high trade elasticity, the losses in terms of exports to high-risk destinations can be very large. At the country level, countries which had focused on exporting products with high trade elasticities and high credit demands will see a larger drop in exports.

E.4.2 Short-run vs. long-run effects

We have written a model in which there are no frictions in adjustment. In particular, in response to a shock to interest rates, firms immediately adjust both their input purchases to reflect the change in prices. Furthermore, prices are also fully flexible, which also removes another possible friction in adjustment. The predictions of the model are therefore only valid for the long-run, as is the case with most trade models. However, in the short-run firms are unable to fully adjust their intermediate purchases as well as the countries to which they sell. In this section, we will take advantage of the findings in [Boehm et al. \(2020\)](#) to compare the short-run vs. long run effects of this trade shock.

In order to motivate this analysis, we can take the gravity equation in equation (E.4) and, after log-linearizing it, we obtain

$$d \log X_{in}^j = d \log X_n^j - \theta^j d \log c_i^j - \theta^j d \log \kappa_{in}^j + \theta^j d \log \Psi_n^j,$$

where the first term represents changes in demand at the destination level, the second term reflects changes in marginal costs in country i , the third term represents the direct effect of changes in trade costs and we can interpret the fourth and final term as a change in the competitive environment.⁴⁰ In our numerical exercise, the term that explains most of the variation (and that does not require general equilibrium forces) is $\theta^j d \log \kappa_{in}^j$. Therefore, conditional on a shock size, the reaction of trade flows will depend on the elasticity. Hence, in order to mimic a world with adjustment frictions, we will consider a sequence of economies with an increasing trade elasticity. This exercise also reflects the findings in [Boehm et al. \(2020\)](#), who find that in response to a variation in trade costs, trade elasticities increase over time.

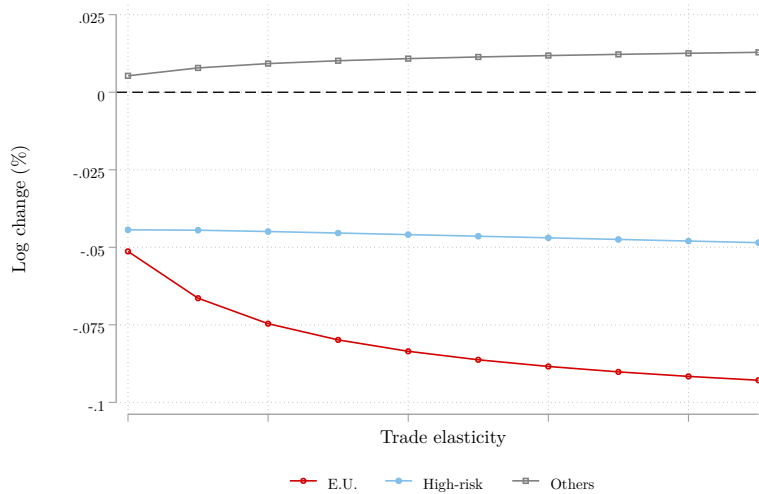
In order to conduct this analysis, we will vary all trade elasticities for tradable costs by multiplying them by a constant inside the unit interval. We fix the size of the shock as in Table VIII and solve the model for each vector of trade elasticities. We can also compute the changes in welfare, measured by the real wage, for different values of the trade

⁴⁰This term contains the changes in marginal costs for all other exporting countries as well as the change in trade costs for exports from all other countries to destination n .

elasticity. For each group of countries - E.U., high-risk and non-E.U. low-risk countries - we compute the average of the changes in the real wage for each value of the trade elasticity while keeping the size of the shock constant. We report the results of this exercise in Figure E.1.

FIGURE E.1: Change in welfare for different trade elasticities

This Figure presents the change in real household income, which is our measure of welfare, for different values of the trade elasticity. For each group of countries - E.U., high-risk and non-E.U. low-risk countries - we compute the average of the changes in real wage while keeping the size of the shock constant. We multiply all dispersions of productivity θ^j (which are also the trade elasticities) of tradable goods by a constant $\alpha \in (0, 1)$ and plot the average change in the real wage for each α .



We start with the evolution of welfare for high-risk countries. We saw in Table IX that most of the variation in the change in welfare is driven by the first-round effect of credit costs. This effect does not depend on the trade elasticity as it does not vary with the ability of firms to adjust. Therefore, changes in welfare for high-risk countries are relatively invariant to the trade elasticity. This is not the case for E.U. countries. In the long-run, which is the last point in the red line in Figure E.1, we recover the effect we presented in Table IX. In the short run, the welfare loss is much smaller and then it increases in absolute value over time as the trade elasticity increases. This evolution is driven by the change in the terms-of-trade effect. In the short run when elasticities are low, most of the welfare effect is driven by the volume-of-trade effect. As trade elasticities increase, global demand for E.U. exports decrease which will imply a larger fall in factor prices. Therefore, the terms-of-trade effect are increase in absolute value as trade elasticities increase.

F Effect on Labor Demand

We have shown that, as the cost of financing for exports to high-risk destinations, firms reduce their exports to these countries. However, we have not discussed the impact of this shock on input demand. As the marginal cost of exporting increases, demand for inputs should decrease as the marginal productivity of inputs becomes smaller than the marginal cost. For instance, we expect that labor demand should decline. As labor demand declines, both wages and the number of workers should decrease at the firm level.⁴¹

F.1 Effect on number of workers

We begin by investigating the impact of Basel III on the number of full-time employees. We aggregate the data at the firm-year level and split firms according to the median share of exports to high-risk destinations in 2013: firms with a share above or equal to the median are the treated firms and firms with a share below the median are the control firms.⁴² To understand the impact of Basel III on the number of full-time employees, we estimate the following regression:

$$\log N_{it} = \alpha_i + \lambda_t + \beta X_{it} + \sum_{m=-3, m \neq -1}^4 \gamma_m \times \mathbf{1}\{i \in \text{Treated}\} \times \mathbf{1}\{m = t - 2014\} + \varepsilon_{it} \quad (\text{F.1})$$

where the outcome variable is the logarithm of the number of full-time employees employed by firm i in year t . We include firm fixed effects, year fixed effects, and a vector of firm controls including firm leverage (total liabilities divided by total assets), the ratio of EBITDA to assets, the ratio of EBITDA to sales, the logarithm of total sales, the ratio of cash and equivalents to total assets, and the ratio of total bank debt to total assets. We also include a vector of firm-level destination controls, including the logarithm of GDP, the logarithm of total exports, the share of imports to GDP, the nominal and real exchange rates, the logarithm of the current account surplus, and the share of the current account to GDP. For each of these variables, we aggregate them to the firm-year level by taking the average across all destinations to which the firm exports using the share of exports to each destination as the weight. The coefficients of interest are the γ_m which represent the effect of Basel III on the number of full-time employees m periods after 2014. We cluster

⁴¹Another possibility is that firms decrease the number of hours. However, decreasing the number of hours is very difficult under Portuguese labor legislation and so firms will either decrease the number of workers or lower wages.

⁴²As before, we exclude firms that only export to high-risk destinations and firms that only export to low-risk destinations. Therefore, we end up with 4,673 firms.

standard errors at the firm level.

There may be substantial heterogeneity in the response of firms to the increase in the cost of credit induced by Basel III. We focus on one dimension - firm size, which we measure using the number of workers in 2013. We split firms according to the median number of workers in 2013. We then estimate equation (F.1) for three samples: (1) using all firms, (2) using firms with a number of workers above the median (large firms), and (3) using firms with a number of workers below the median (small firms). We present the results of this analysis in Figure F.1.

We find that there is no statistically significant effect on the average number of workers employed by exporters following the implementation of Basel III. All of the coefficients following 2014 are negative but are not statistically significant. However, there is heterogeneity across firms. We find that, if we focus on large firms (firms with a large number of workers), there is no evidence that treated firms reduce the number of employees. In contrast, smaller treated firms sharply reduce the number of employees. In 2018, treated firms reduce the number of employees by 5.4 percent which represents a decrease of approximately one worker relative to the average number of workers in 2013.

F.2 Effect on wages

We now turn to the analysis of the effect of Basel III on monthly wages. Using our worker-level labor data and splitting firms into treated and control groups as before, we estimate the following regression:

$$\log w_{ijt} = \alpha_i + \lambda_t + \theta_{\text{occ}} + \beta X_{it} + \delta W_{jt} + \sum_{m=-3, m \neq -1}^4 \gamma_m \times \mathbf{1}\{i \in \text{Treated}\} \times \mathbf{1}\{m = t - 2014\} + \varepsilon_{it} \quad (\text{F.2})$$

where the outcome variable is the logarithm of the monthly wage of worker j in firm i in year t . We include firm fixed effects, year fixed effects, and occupation fixed effects.⁴³ We also include a vector of firm-level controls as in equation (12). We include a vector of worker controls, which contains a gender fixed effect, the logarithm of the worker's age, and the logarithm of the worker's tenure at the current firm. The coefficients of interest are the γ_m which represent the effect of Basel III on the wage m periods after 2014. We cluster standard errors at the firm level. As before, we estimate (F.2) for three samples: (1) using all firms, (2) using firms with a number of workers above the median (large firms),

⁴³In our worker-level labor data, we can identify the occupation of each worker for every observation. We observe 398 different occupations in our data. The inclusion of these fixed effects allows us to account for possible differences in the skill level across workers.

and (3) using firms with a number of workers below the median (small firms). We present the results in Figure [F.2](#).

We find that, in response to Basel III, treated firms decrease their wages. However, this decrease only starts in 2016 unlike the decline in number of employees which starts earlier (around 2015). The delay in the wage decline is likely driven by the fact that firms either need time to negotiate wages with incumbent workers, or need time to replace incumbent workers with similar workers with lower wages. In 2016, workers in treated firms observe a decline of 2.2 percent in their monthly wage. The wage seems to recover slightly towards the end of our sample as the decline in 2018 is only 1.4 percent. Therefore, for the average treated firm, there seems to be a decline in wages paid to workers as well as a decline in the number of workers. We find that most of the decline in wages we observe for the average treated firm is driven by large firms, with the exception of an outlier in 2015. For example, in 2016, workers in large treated firms experience a decline of 2.4 percent in their wage, which is basically the same decline we observe for the average treated firm. This decline is also somewhat persistent following 2016, with a slight recovery in 2018. Therefore, the response of firms to an increase in the cost of credit depends on the size of the firm: large firms reduce wages while small firms reduce the number of workers.

F.2.1 Heterogeneity across workers

We have documented that, in response to the increase in the cost of credit introduced by Basel III, treated firms decrease wages. However, this decrease masks some heterogeneity across workers. In particular, we are interested in understanding which workers bear the brunt of this decline. To understand this, we start by comparing the evolution of wages of managers with that of other workers. In our data, firms report the hierarchical position of each worker within the firm.⁴⁴ We therefore estimate equation [\(F.2\)](#) in three samples: (1) using all observations, (2) using observations for managers, and (3) using observations for non-managers. We are also interesting in understanding which occupations face a larger decline in wages. For each occupation, we compute the average wage in 2013. We then classify occupation in two groups: occupations with an average wage above or equal to the median in 2013 are classified as high- wage occupations, and the remaining occupations are classified as low-wage occupations. We then estimate equation [\(13\)](#) in three

⁴⁴Firms must assign each worker to one of seven levels: top manager, medium-level manager, low-level manager, highly qualified worker, qualified worker, semi-qualified worker, unqualified worker, and intern. We define managers as workers assigned to any of the first three levels. This classification is identical to the one used in [Caliendo et al. \(2020\)](#).

samples: (1) using all observations, (2) using observations for workers in low-wage occupations, and (3) using observations for workers in high-wage occupations. We present the result of this analysis in Figure F.3.

In panel (a), we find that all of the decline in wages is driven by workers who do not occupy managerial positions. In 2018, non-managers in treated firms experience a decline of 1.5 percent in their wage, while managers in treated firms do not exhibit any change in their wage relative to 2013. There are two possible causes for this asymmetry. One possibility is that, in response to the Basel III shock, treated firms wish to decrease labor costs for unskilled workers but not for skilled workers. Another possibility is related to renegotiation costs. Firms may wish to lower wages for all workers, but it might be costlier to lower wages for managers (either because renegotiating with managers is harder or because replacing incumbent managers with new managers with lower wages is harder). In panel (b), we find that workers in occupations that have lower wages in 2013 exhibit a larger decline in wages following the implementation of Basel III. This finding is consistent with the idea that treated firms want to reduce by more the wages of unskilled workers, when compared to more skilled workers.

FIGURE F.1: Effect on number of full-time employees

This figure presents the results of estimating equation (F.1) on a sample of 4,673 firms between 2011 and 2018. The outcome variable is the logarithm of the number of full-time employees for each firm and year. We compare two groups of firms based on the share of exports to high-risk destinations in 2013: firms with a share above or equal to the median are the treated firms and firms with a share below the median are the control firms. We include firm fixed effects, year fixed effects, and a vector of firm controls including firm leverage (total liabilities divided by total assets), the ratio of EBITDA to assets, the ratio of EBITDA to sales, the logarithm of total sales, the ratio of cash and equivalents to total assets, and the ratio of total bank debt to total assets. We also include a vector of firm-level destination controls, including the logarithm of GDP, the logarithm of total exports, the share of imports to GDP, the nominal and real exchange rates, the logarithm of the current account surplus, and the share of the current account to GDP. For each of these variables, we aggregate them to the firm-year level by taking the average across all destinations to which the firm exports using the share of exports to each destination as the weight. We use the year 2013 as the base year and present average treatment effects for treated firms relative to 2013. We further divide firms according to the number of workers in 2013, using the median in 2013. We estimate equation (12) for three samples: (1) using all firms, (2) using firms with a number of workers above the median, and (3) using firms with number of workers below the median. We cluster errors at the firm-level and present 95 percent confidence intervals.

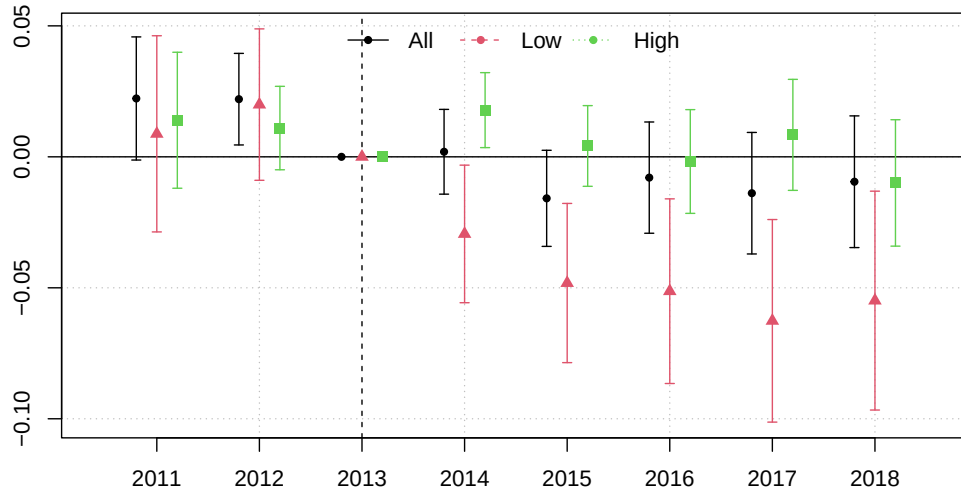


FIGURE F.2: Effect on wages

This figure presents the results of estimating equation (F.2) on a sample of 600,308 workers and 4,673 firms between 2011 and 2018. The outcome variable is the logarithm of the monthly wage for each worker, firm, and year. We compare two groups of firms based on the share of exports to high-risk destinations in 2013: firms with a share above or equal to the median are the treated firms and firms with a share below the median are the control firms. We include firm fixed effects, year fixed effects, occupation fixed effects, and a vector of firm controls including firm leverage (total liabilities divided by total assets), the ratio of EBITDA to assets, the ratio of EBITDA to sales, the logarithm of total sales, the ratio of cash and equivalents to total assets, and the ratio of total bank debt to total assets. We also include a vector of firm-level destination controls, including the logarithm of GDP, the logarithm of total exports, the share of imports to GDP, the nominal and real exchange rates, the logarithm of the current account surplus, and the share of the current account to GDP. For each of these variables, we aggregate them to the firm-year level by taking the average across all destinations to which the firm exports using the share of exports to each destination as the weight. We also include worker controls: a gender fixed effect, the logarithm of the worker's age, and the logarithm of the worker's tenure. We use the year 2013 as the base year and present average treatment effects for treated firms relative to 2013. We further divide firms according to the number of workers in 2013, using the median in 2013. We estimate equation (12) for three samples: (1) using all firms, (2) using firms with a number of workers above the median, and (3) using firms with number of workers below the median. We cluster errors at the firm-level and present 95 percent confidence intervals.

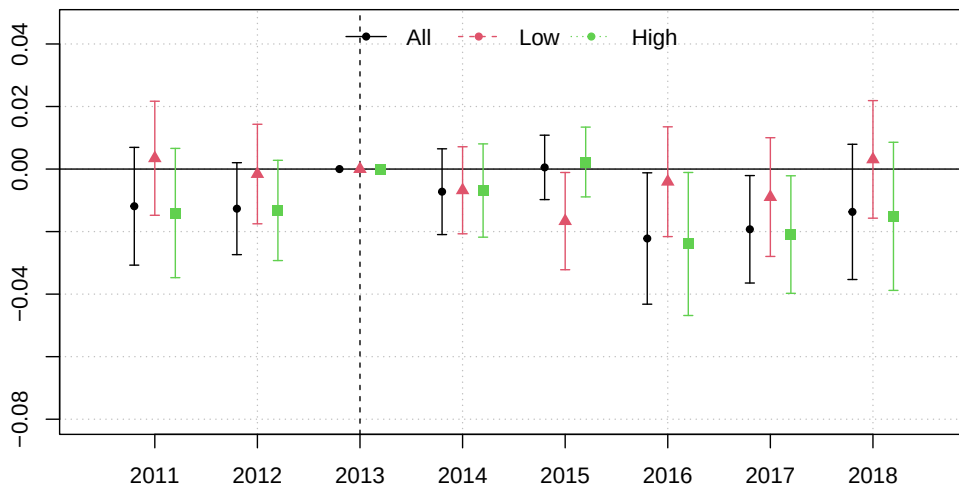
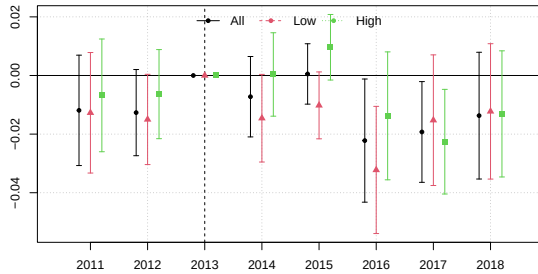


FIGURE F.3: Effect on wages - heterogeneity across workers

This figure presents the results of estimating equation (F.2) on a sample of 600,308 workers and 4,673 firms between 2011 and 2018. The outcome variable is the logarithm of the monthly wage for each worker, firm, and year. We compare two groups of firms based on the share of exports to high-risk destinations in 2013: firms with a share above or equal to the median are the treated firms and firms with a share below the median are the control firms. We include firm fixed effects, year fixed effects, occupation fixed effects, and a vector of firm controls including firm leverage (total liabilities divided by total assets), the ratio of EBITDA to assets, the ratio of EBITDA to sales, the logarithm of total sales, the ratio of cash and equivalents to total assets, and the ratio of total bank debt to total assets. We also include a vector of firm-level destination controls, including the logarithm of GDP, the logarithm of total exports, the share of imports to GDP, the nominal and real exchange rates, the logarithm of the current account surplus, and the share of the current account to GDP. For each of these variables, we aggregate them to the firm-year level by taking the average across all destinations to which the firm exports using the share of exports to each destination as the weight. We also include worker controls: a gender fixed effect, the logarithm of the worker's age, and the logarithm of the worker's tenure. We use the year 2013 as the base year and present average treatment effects for treated firms relative to 2013. We divide workers into two groups according to their hierarchical position: managers vs. non-managers. In panel (a), we estimate equation (13) for three samples: (1) using all observations, (2) using observations for managers, and (3) using observations for non-managers. For each occupation, we compute the average wage in 2013 and then sort occupations into two groups: occupations with an average wage above or equal to the median are classified as high-wage occupations, and occupations with an average wage below the median are classified as low-wage occupations. In panel (b), we estimate equation (13) for three samples: (1) using all observations, (2) using all observations for workers in low-wage occupations, and (3) using all observations for workers in high-wage occupations. We cluster errors at the firm-level and present 95 percent confidence intervals.

(a) Hierarchical position



(b) Occupation wage

